
ScoreMatchingRiesz: Score Matching for Debiased Machine Learning and Policy Path Estimation

Masahiro Kato¹

Abstract

We propose *ScoreMatchingRiesz*, a family of Riesz representer estimators based on score matching. The Riesz representer is a key nuisance component in debiased machine learning, enabling $\sqrt{n}$ -consistent and asymptotically efficient estimation of causal and structural targets via Neyman-orthogonal scores. We formulate Riesz representer estimation as a score estimation problem. This perspective stabilizes representer estimation by allowing us to leverage denoising score matching and telescoping density ratio estimation. We also introduce the policy path, a parameter that captures how policy effects evolve under continuous treatments. We show that the policy path can be estimated via score matching by smoothly connecting average marginal effect (AME) and average policy effect (APE) estimation, which improves the interpretability of policy effects. The replication code is available in the `genriesz` library.

1. Introduction

Estimating the Riesz representer is a central problem in debiased machine learning (DML) for causal inference and structural parameter estimation (Chernozhukov et al., 2024a). For example, in average treatment effect (ATE) estimation, the Riesz representer corresponds to inverse propensity score weights, and accurate estimation of this representer is essential for accurate ATE estimation. *Score matching* refers to methods for estimating the score, the derivative of a log density with respect to a variable. Score matching has recently attracted renewed attention through the diffusion model literature (Song & Ermon, 2020).

Riesz representer estimation and score matching are closely

related, but leveraging this link for DML requires more than simply importing existing score estimators. For example, in average marginal effect (AME) estimation, the Riesz representer coincides with the (negative) score of the joint density of (D, Z) with respect to the treatment coordinate D given covariates Z . Moreover, Hyvärinen score matching (Hyvärinen, 2005) yields an objective that is closely related to Riesz regression (Chernozhukov et al., 2021). However, Hyvärinen-type objectives can be numerically challenging in high dimensions because they involve divergence computations. In applications where the representer is written in terms of density ratios (Kato, 2025a; 2026a), one may instead turn to density ratio estimation (DRE), but flexible DRE is known to suffer from severe overfitting (Kato & Teshima, 2021; Rhodes et al., 2020). While denoising score matching (DSM) (Vincent, 2011) and DRE- ∞ (Choi et al., 2022) have been developed to stabilize score and ratio estimation, their use as *Riesz representer estimators* within Neyman orthogonal scores calls for an explicit construction and an analysis.

Motivated by these relationships, this paper develops score matching-based methods for Riesz representer estimation, which we call *ScoreMatchingRiesz*. The key idea is to use score models as a Riesz representer estimation module: we construct the Riesz representer either directly from a score field or via density ratios recovered by integrating a score along a bridge of intermediate distributions. Our framework involves two types of scores. The *data score* is the derivative of the log density with respect to the data variable, and it is estimated by diffusion-style score matching. The *time score* is the derivative of the log density along a continuum of intermediate distributions, and it is estimated via infinitesimal classification-based score matching (the continuum limit underlying DRE- ∞). We refer to *ScoreMatchingRiesz* designed for data-score estimation as *Data-ScoreMatchingRiesz* and that designed for time-score estimation as *Time-ScoreMatchingRiesz*.

In AME estimation, the Riesz representer can be estimated directly from the data score. In other applications such as APE and ATE estimation, however, estimating the Riesz representer requires estimating density ratios, and time-score estimation provides a general way to recover these

¹Mizuho-DL Financial Technology Co., Ltd., Tokyo, Japan. Correspondence to: Masahiro Kato <mkato-csecon@g.ecc.u-tokyo.ac.jp>.

Table 1. Specific forms of ScoreMatchingRiesz (SMR).

	ATE	AME	APE with a known pushforward policy	APE with a non-pushforward policy
Data-SMR		✓	✓	
Time-SMR	✓	✓	✓	✓

ratios through a bridge between endpoint distributions. In some restricted settings, the data-score construction is sufficient on its own. For example, in APE estimation under a known pushforward policy, the relevant density ratio can be expressed as an integral of the data score along a known path, which yields the representer without explicitly learning an endpoint-to-endpoint ratio. Joint training of data and time scores can further stabilize the estimation process across these regimes. See Table 1.

This score-based representer construction also provides a concrete link between AME and APE through a continuum of policy effects. In this study, we define the *policy path* as a parameter of interest and propose an estimator. In shift-policy settings, the APE representer can be constructed from the AME representer by integrating the score along the policy path. Using this relationship, we estimate intermediate policy effects as the policy shifts continuously.

In summary, our contributions are as follows:

- We develop *ScoreMatchingRiesz*, a score matching-based approach that estimates Riesz representers from data scores and time scores, with two named estimators: Data-ScoreMatchingRiesz and Time-ScoreMatchingRiesz.
- We adapt the generalized Riesz regression perspective based on Bregman divergence minimization (Kato, 2025a; 2026d) to the time-score construction used in ScoreMatchingRiesz.
- We define the policy path as a parameter of interest that bridges AME and APE, and we propose an estimation method based on score integration along the path.
- We establish convergence rate results for the proposed Riesz representer estimators, which are sufficient for DML via Neyman orthogonal scores.

There are additional discussions in the appendices due to space constraints. For example, Appendix G gives conditions under which the AME limit and the APE path are connected through known pushforward policies. Appendix Q provides an application to local projection (Jordà, 2005).

Related work. Constructing (asymptotically) efficient estimators is a central task in causal inference (van der Vaart, 1998), and various unified perspectives on this task have been established. Among them, the DML framework formulates the problem through Neyman orthogonal score equations together with cross-fitting (Chernozhukov et al., 2018). Various methods for Riesz representer estimation have been proposed, including Riesz regression

(Chernozhukov et al., 2021; 2022a; 2024b), nearest neighbor matching (Kato, 2025c; Lin et al., 2023), and covariate balancing (Zhao, 2019; Bruns-Smith et al., 2025).

The goal of DRE is to estimate the ratio between two probability density functions. Many methods have been proposed for DRE (Sugiyama et al., 2012), but flexible DRE is known to suffer from severe overfitting, called train loss hacking (Kato & Teshima, 2021) or the density-chasm (Rhodes et al., 2020). One approach to mitigating this overfitting is to avoid estimating a single log density ratio in one step and instead introduce a sequence of intermediate density ratios (Rhodes et al., 2020). Choi et al. (2022) proposes an infinitesimal classification approach, which takes a continuum limit of telescoping ratios. We use this idea through Time-ScoreMatchingRiesz.

As we and prior work note (Kato, 2025a; 2026d; Ben-Michael et al., 2021), Riesz representer estimation admits complementary views via score matching and DRE. As a consequence, the instability issues known for Hyvärinen score matching and the overfitting issues documented for flexible DRE can also arise in Riesz representer estimation. We make these connections operational by constructing Riesz representers from data scores and time scores and by providing convergence guarantees in the form needed for DML. In particular, the Bregman-divergence formulation in Section 6 uses the generalized Riesz regression perspective of Kato (2025a; 2026d); the new part here is its adaptation to score-based representer construction and to the downstream error propagation for Neyman orthogonal estimation. Also see Kato (2026b) for the relationship for covariate balancing.

For detailed related work, see Appendix A.

2. Setup

Let $W = (X, Y)$ consist of a regressor $X \in \mathcal{X} \subseteq \mathbb{R}^d$ and a scalar outcome $Y \in \mathcal{Y} \subseteq \mathbb{R}$. Assume $(X, Y) \sim P_0$, and define the regression function $\gamma_0(x) := \mathbb{E}[Y | X = x]$. We observe $\{(X_i, Y_i)\}_{i=1}^n$, where (X_i, Y_i) are i.i.d. copies of (X, Y) . Our target parameter is

$$\theta_0 := \mathbb{E}[m(W, \gamma_0)],$$

where $m(W, \gamma)$ is a functional of the data and a regression function $\gamma: \mathcal{X} \rightarrow \mathcal{Y}$. We assume that the map $\gamma \mapsto \mathbb{E}[m(W, \gamma)]$ is *linear and continuous* on $L_2(P_X)$ (the L_2 space under the marginal law of X).

2.1. Riesz representer and Neyman orthogonal score

By the Riesz representation theorem, there exists a function $\alpha_0 \in L_2(P_X)$ such that, for all $\gamma \in L_2(P_X)$,

$$\mathbb{E}[m(W, \gamma)] = \mathbb{E}[\alpha_0(X)\gamma(X)]$$

holds. We call α_0 the *Riesz representer* associated with m .

Let $\eta_0 := (\gamma_0, \alpha_0)$ denote the nuisance parameters. For a candidate pair $\eta = (\gamma, \alpha)$ and scalar θ , let us define the Neyman orthogonal score

$$\psi(W; \eta, \theta) := m(W, \gamma) + \alpha(X)(Y - \gamma(X)) - \theta.$$

Then $\mathbb{E}[\psi(W; \eta_0, \theta_0)] = 0$ and the moment condition is orthogonal to first-order perturbations of η at η_0 under regularity conditions (Chernozhukov et al., 2018).

2.2. Estimation of the parameter of interest

An estimator $\hat{\theta}$ is obtained by plugging in nuisance estimates $\hat{\eta} = (\hat{\gamma}, \hat{\alpha})$ and solving $\frac{1}{n} \sum_{i=1}^n \psi(W_i; \hat{\eta}, \hat{\theta}) = 0$; equivalently, the estimator is given by

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n \left(m(W_i, \hat{\gamma}) + \hat{\alpha}(X_i)(Y_i - \hat{\gamma}(X_i)) \right).$$

Assume that $\hat{\eta}$ satisfies the Donsker condition or is constructed via cross-fitting. Then, under suitable convergence rate conditions on $\hat{\eta}$, we have $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, V)$, where $V > 0$ is the efficient asymptotic variance.

Notation and assumptions. For simplicity, we assume that the components of X are continuous, except in specific cases such as the treatment D in ATE estimation. We also assume that each continuous random variable R admits a probability density function $p_0(r)$ under P_0 . We write $R \sim p_0$. Expectations are taken under P_0 unless stated otherwise; when needed, we write $\mathbb{E}_{R \sim p}[\cdot]$ for R with density p . We use t , δ , and σ to index the bridge, policy shift, and Gaussian noise level, respectively. The data score and time score are defined in Section 3.1.

2.3. Applications

We illustrate how the Riesz representer appears in applications. Also see Appendix E.

AME. Let $X = (D, Z)$ with continuous $D \in [-1, 1]$. The AME is

$$\theta_0^{\text{AME}} := \mathbb{E}[\partial_d \gamma_0(D, Z)], \quad m^{\text{AME}}(W, \gamma) := \partial_d \gamma(D, Z).$$

Under standard integration-by-parts conditions (including suitable boundary behavior), the Riesz representer is the negative score of the joint density of (D, Z) with respect to d and given as

$$\alpha_0^{\text{AME}}(D, Z) := -\partial_d \log p_0(d, Z)|_{d=D}.$$

APE. Let P_1 and P_{-1} be two counterfactual distributions of X with densities $p_1(x)$ and $p_{-1}(x)$, both absolutely continuous with respect to P_0 (density p_0). The APE is

$$\theta_0^{\text{APE}} := \mathbb{E}_{X \sim p_1}[\gamma_0(X)] - \mathbb{E}_{X \sim p_{-1}}[\gamma_0(X)].$$

This can be written as $\theta_0^{\text{APE}} = \mathbb{E}[m^{\text{APE}}(W, \gamma_0)]$ with $m^{\text{APE}}(W, \gamma) := \gamma(X) \frac{p_1(X) - p_{-1}(X)}{p_0(X)}$. Hence, the Riesz representer is given as

$$\alpha_0^{\text{APE}}(X) := \frac{p_1(X) - p_{-1}(X)}{p_0(X)}.$$

3. Recaps of Riesz Regression, Score Matching, and DRE- ∞

This section reviews Riesz regression (Chernozhukov et al., 2021), score matching (Song et al., 2021), and DRE- ∞ (Choi et al., 2022). Score matching can be carried out via Hyvärinen score matching or denoising score matching (DSM). In AME estimation, the Riesz regression objective for the treatment-direction representer coincides with a one-coordinate Hyvärinen score-matching objective and, in certain linear models, with a least-squares importance fitting (LSIF)-type objective.

3.1. Definitions of Scores

Throughout this subsection, let $X \in \mathbb{R}^d$ be a continuously distributed random vector with density $p_0(x)$ under P_0 . We define the data score of p_0 as

$$(\text{Data score}) \quad s_0^{\text{data}}(x) := \nabla_x \log p_0(x) \in \mathbb{R}^d.$$

The Riesz representer in AME estimation directly corresponds to the data score as $\alpha_0^{\text{AME}}(D, Z) = -\partial_d \log p_0(D, Z)$.

We introduce a parametric family $\{p_t(x)\}_{t \in \mathcal{T}}$ such that $p_t(x) = p_0(x)$ at $t = 0$. This family is specified by the researcher based on applications. We refer to the parameter t as “time.” Using this family, we define the time score as

$$(\text{Time score}) \quad s_t^{\text{time}}(x) := \partial_t \log p_t(x) \in \mathbb{R}.$$

3.2. One-Coordinate Hyvärinen Score Matching, Riesz Regression, and LSIF in AME Estimation

We first consider the estimation of the data score s_0^{data} . Let $s^{\text{data}}: \mathcal{X} \rightarrow \mathbb{R}^d$ be a candidate score function (e.g., a neural network s_θ^{data}). A natural population criterion is the mean squared error (MSE) between s and the true score s_0^{data} : $\mathcal{R}^\dagger(s^{\text{data}}) := \frac{1}{2} \mathbb{E}_{X \sim p_0} [\|s^{\text{data}}(X) - s_0^{\text{data}}(X)\|_2^2]$. This objective is not directly computable because s_0^{data} is unknown. However, under standard integration-by-parts conditions, minimizing $\mathcal{R}^\dagger$ is equivalent to minimizing

$$\mathcal{R}^{\text{HSM}}(s^{\text{data}}) := \mathbb{E}_{X \sim p_0} \left[\frac{1}{2} \|s^{\text{data}}(X)\|_2^2 + \nabla_x \cdot s^{\text{data}}(X) \right], \quad (1)$$

where $\nabla_x \cdot s^{\text{data}}(x) := \sum_{j=1}^d \partial_{x_j} s_j^{\text{data}}(x)$. This estimation approach is the multivariate Hyvärinen score-matching

objective. For AME, the corresponding Riesz regression objective uses the treatment-coordinate component and is therefore a one-coordinate Hyvärinen score-matching objective; see Appendix B.

3.3. DSM

In high dimensions, directly optimizing Eq. (1) can be numerically challenging because it involves the divergence $\nabla_x \cdot s_\theta(X)$, which requires Jacobian-trace computations. A widely used alternative is DSM (Vincent, 2011), which replaces the (possibly rough) target score s_0^{data} by the score of a *Gaussian-smoothed* density. We explain the details in Section 4. In this study, we refer to DSM-based Riesz representer estimation as Data-ScoreMatchingRiesz.

3.4. DRE- ∞

Let $q(x)$ and $p(x)$ be two densities on $\mathcal{X}$, and assume q is absolutely continuous with respect to p on $\mathcal{X}$. Define the density ratio as $r(x) := q(x)/p(x)$. Assume we can draw i.i.d. samples from q and p .

Bridge distributions and telescoping ratios. Following Choi et al. (2022), construct a continuum of *bridge* distributions $\{p_t\}_{t \in [0,1]}$ by sampling

$$X_t := \beta^{(1)}(t)X_0 + \beta^{(2)}(t)X_1,$$

where $X_0 \sim q$, $X_1 \sim p$ are independent, and $\beta^{(1)}, \beta^{(2)} : [0,1] \rightarrow [0,1]$ are C^2 functions satisfying $\beta^{(1)}(0) = 1$, $\beta^{(2)}(0) = 0$, $\beta^{(1)}(1) = 0$, $\beta^{(2)}(1) = 1$. Let p_t denote the law of X_t (and write $p_t(x)$ for its density when it exists).

Using intermediate ratios, we have for any integer $T \geq 1$, $r(x) = \frac{q(x)}{p(x)} = \prod_{k=1}^T \frac{p_{(k-1)/T}(x)}{p_{k/T}(x)}$. Taking logs gives $\log r(x) = \sum_{k=1}^T \log \frac{p_{(k-1)/T}(x)}{p_{k/T}(x)}$. In the continuum limit, the following result holds:

Proposition 3.1 (Continuum limit of telescoping ratios; cf. Proposition 1 in Choi et al. (2022)). *As $T \rightarrow \infty$, we have*

$$\log r(x) = \int_0^1 \partial_t \log p_t(x) dt = - \int_0^1 \partial_t \log p_t(x) dt.$$

The time score is $s_t^{\text{time}}(x) = \partial_t \log p_t(x)$.

Time score matching objective. Let $s_{\theta,t}^{\text{time}}(x)$ be a time score model. A natural population loss is $\mathcal{R}^\dagger(s_{\theta,t}^{\text{time}}) := \frac{1}{2} \int_0^1 \mathbb{E}_{X_t \sim p_t} \left[\lambda(t) \left(\partial_t \log p_t(X_t) - s_{\theta,t}^{\text{time}}(X_t) \right)^2 \right] dt$, where $\lambda : [0,1] \rightarrow \mathbb{R}_+$ is a weighting function. The true time score is unknown, but Choi et al. (2022) shows that (up to an additive constant that does not depend on θ) this objective is equivalent to an integration-by-parts objective:

$$\mathcal{R}(s_{\theta}^{\text{time}}) := \mathbb{E}_{X_0 \sim q} [\lambda(0) s_{\theta,0}^{\text{time}}(X_0)] - \mathbb{E}_{X_1 \sim p} [\lambda(1) s_{\theta,1}^{\text{time}}(X_1)] + \int_0^1 \mathbb{E}_{X_t \sim p_t} \left[\partial_t (\lambda(t) s_{\theta,t}^{\text{time}}(X_t)) + \frac{1}{2} \lambda(t) s_{\theta,t}^{\text{time}}(X_t)^2 \right] dt. \quad (2)$$

$$\mathbb{E}_{X_0 \sim q} [\lambda(0) s_{\theta,0}^{\text{time}}(X_0)] - \mathbb{E}_{X_1 \sim p} [\lambda(1) s_{\theta,1}^{\text{time}}(X_1)] + \int_0^1 \mathbb{E}_{X_t \sim p_t} \left[\partial_t (\lambda(t) s_{\theta,t}^{\text{time}}(X_t)) + \frac{1}{2} \lambda(t) s_{\theta,t}^{\text{time}}(X_t)^2 \right] dt.$$

In the integral term, $\partial_t s_{\theta,t}^{\text{time}}(x)$ denotes the *partial* derivative with respect to t while holding x fixed.

Recovering the density ratio. After fitting $s_{\theta,t}^{\text{time}}$, the ratio can be estimated by numerical integration as $\widehat{\log r}(x) := \int_0^1 s_{\theta,t}^{\text{time}}(x) dt$ and $\widehat{r}(x) := \exp(\widehat{\log r}(x))$.

4. Data-ScoreMatchingRiesz

We first define Data-ScoreMatchingRiesz, which is an application of DSM. Let $\sigma > 0$ and define a noising kernel

$$q_\sigma(\tilde{x} | x) = \mathcal{N}(\tilde{x}; x, \sigma^2 I_d), \quad \tilde{X} = X + \sigma \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I_d).$$

Let p_σ denote the marginal density of $\tilde{X}$, i.e. the convolution $p_\sigma = p_0 * \mathcal{N}(0, \sigma^2 I_d)$. Data-ScoreMatchingRiesz fits a model $s_{\theta,\sigma}^{\text{data}}(\tilde{x})$ by minimizing $\mathcal{R}^{\text{DSM}\dagger}(s_{\theta,\sigma}^{\text{data}}) := \mathbb{E}_{\sigma \sim \pi} \mathbb{E}_{X \sim p_0} \mathbb{E}_{\tilde{X} \sim q_\sigma(\cdot | X)} [\lambda(\sigma) \| s_{\theta,\sigma}^{\text{data}}(\tilde{X}) - \nabla_{\tilde{x}} \log q_\sigma(\tilde{X} | X) \|_2^2]$, where π is a distribution over noise levels and $\lambda(\sigma) \geq 0$ is a weighting function. For Gaussian noise, $\nabla_{\tilde{x}} \log q_\sigma(\tilde{x} | x) = -\frac{\tilde{x}-x}{\sigma^2} = -\frac{\varepsilon}{\sigma}$ holds. Hence, an equivalent feasible form of $\mathcal{R}^{\text{DSM}\dagger}(s_{\theta,\sigma}^{\text{data}})$ is

$$\mathcal{R}^{\text{DSM}}(s_{\theta,\sigma}^{\text{data}}) = \mathbb{E}_{\sigma \sim \pi} \mathbb{E}_{X \sim p_0} \mathbb{E}_{\varepsilon \sim \mathcal{N}(0, I_d)} \left[\lambda(\sigma) \| s_{\theta,\sigma}^{\text{data}}(X + \sigma \varepsilon) + \varepsilon / \sigma \|_2^2 \right].$$

It is known that for each fixed σ , the population minimizer θ^* satisfies $s_{\theta^*,\sigma}^{\text{data}}(\cdot) := \nabla_{\tilde{x}} \log p_\sigma(\cdot)$.

This is the formulation most commonly used in diffusion models (Song et al., 2021): one trains a single network on a continuum of noise levels σ , which can be interpreted as a “time” variable, and then evaluates (or extrapolates) the score near the small-noise regime.

5. Time-ScoreMatchingRiesz

The (time) score matching approach above extends directly to Riesz representer estimation when the representer can be expressed through density ratios.

Remark (Densities and endpoint degeneracy). *We use p_t to denote the law of a bridge sample induced by an explicit sampling map. When this law is absolutely continuous, we write $p_t(x)$ for its density and define the time score $\partial_t \log p_t(x)$. Some bridge constructions can yield degenerate endpoint laws. This does not prevent evaluation of expectations, but it motivates either interpreting endpoint terms as limits or downweighting endpoints by choosing λ with $\lambda(0) = \lambda(1) = 0$ or sampling t from $[\varepsilon, 1 - \varepsilon]$. The*

rate statement in Theorem 8.2 is stated for $\lambda_{\min} > 0$; if end-point downweighting is used, the same statement should be interpreted on a trimmed interval or under a lower bound away from the endpoints.

5.1. AME estimation

In AME estimation with continuous D , the Riesz representer is $\alpha_0^{\text{AME}}(D, Z) = -\partial_d \log p_0(d, Z)|_{d=D}$. This Riesz representer can be interpreted as a time score.

Time score interpretation. For intuition, consider the affine bridge $D_t = \beta^{(1)}(t) + \beta^{(2)}(t)D$ with $\beta^{(1)}(0) = 0$, $\beta^{(2)}(0) = 1$, and derivatives $\partial_t \beta^{(1)}(0) = 1$, $\partial_t \beta^{(2)}(0) = 0$. Let $p_t(d, z)$ be the density of (D_t, Z) . A direct change-of-variables calculation yields, for t where a density exists,

$$\begin{aligned} \partial_t \log p_t(d, z) = & -\frac{\partial_t \beta^{(2)}(t)}{\beta^{(2)}(t)} \\ & - \frac{\partial_t \beta^{(1)}(t) + u \partial_t \beta^{(2)}(t)}{\beta^{(2)}(t)} \partial_u \log p_0(u, z) \Big|_{u=\frac{d-\beta^{(1)}(t)}{\beta^{(2)}(t)}}. \end{aligned}$$

Evaluating at $t = 0$ gives $\partial_t \log p_t(d, z)|_{t=0} = -\partial_d \log p_0(d, z)$, so the AME representer can be viewed as a time score at $t = 0$ for this local translation-like bridge. This observation connects AME estimation to the time-score framework used for ratio-based representers.

5.2. APE estimation

In APE estimation, the Riesz representer is the difference of two density ratios: $\alpha_0^{\text{APE}}(x) = \frac{p_1(x)}{p_0(x)} - \frac{p_{-1}(x)}{p_0(x)}$. Thus it suffices to estimate p_1/p_0 and p_{-1}/p_0 .

Bridge construction and time score training. Construct a bridge family $\{p_t\}_{t \in [-1, 1]}$ for X using independent draws $X_0 \sim p_0$, $X_1 \sim p_1$, $X_{-1} \sim p_{-1}$ and the same style of interpolation as in the ATE case:

$$X_t := \begin{cases} \beta^{(1)}(t)X_0 + \beta^{(2)}(t)X_1, & t \in [0, 1], \\ \beta^{(1)}(-t)X_0 + \beta^{(2)}(-t)X_{-1}, & t \in [-1, 0]. \end{cases}$$

We train $s_{\theta, t}^{\text{time}}(x)$ by minimizing

$$\begin{aligned} \mathcal{R}^{\text{APE}}(s_{\theta}^{\text{time}}) := & \mathbb{E}_{X_{-1} \sim p_{-1}} [\lambda(-1)s_{\theta, -1}^{\text{time}}(X_{-1})] \\ & - \mathbb{E}_{X_1 \sim p_1} [\lambda(1)s_{\theta, 1}^{\text{time}}(X_1)] \\ & + \int_{-1}^1 \mathbb{E}_{X_t \sim p_t} \left[\partial_t (\lambda(t)s_{\theta, t}^{\text{time}}(X_t)) + \frac{1}{2} \lambda(t)s_{\theta, t}^{\text{time}}(X_t)^2 \right] dt. \end{aligned}$$

Constructing $\hat{\alpha}^{\text{APE}}$. After training, we obtain $\widehat{\log \frac{p_1}{p_0}}(x) := \int_0^1 s_{\theta, t}^{\text{time}}(x) dt$ and $\widehat{\log \frac{p_{-1}}{p_0}}(x) := \int_0^{-1} s_{\theta, t}^{\text{time}}(x) dt$. Then, we estimate the Riesz representer as $\hat{\alpha}^{\text{APE}}(x) := \exp \left(\widehat{\log \frac{p_1}{p_0}}(x) \right) - \exp \left(\widehat{\log \frac{p_{-1}}{p_0}}(x) \right)$.

5.3. Joint Training of the Scores

To stabilize training, we jointly train the data score while training the time score, following the suggestion in Choi et al. (2022), in a multitask learning manner. For details, see Appendix D.

5.4. Estimation of the Parameter of Interest

The procedure for estimating the parameter of interest is as follows: (1) estimate the nuisance parameters $\eta_0 = (\gamma_0, \alpha_0)$; (2) plug in the estimates $\hat{\eta} = (\hat{\gamma}, \hat{\alpha})$ into the Neyman score and solve for $\hat{\theta}$ from $\frac{1}{n} \sum_{i=1}^n \psi(W_i; \hat{\eta}, \hat{\theta}) = 0$.

If the estimators of the nuisance parameters do not satisfy the Donsker condition, we employ cross-fitting to control the empirical process term (Klaassen, 1987; Zheng & van der Laan, 2011; Chernozhukov et al., 2018).

6. Generalization of ScoreMatchingRiesz

ScoreMatchingRiesz admits a generalized formulation based on Bregman divergences, which makes the choice of the loss and the link explicit. The Bregman-Riesz perspective was developed in Kato (2025a) and refined in Kato (2026d). We do not claim this perspective itself as new. The role of this section is to adapt the loss-link viewpoint to ScoreMatchingRiesz. Due to space constraints, we focus on Time-ScoreMatchingRiesz; the same argument applies to Data-ScoreMatchingRiesz.

6.1. Generalization via Bregman Divergence

Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable and strictly convex, and let $D_g(u||v) := g(u) - g(v) - g'(v)(u - v)$ denote the Bregman divergence. For the bridge $\{p_t\}_{t \in [0, 1]}$ with time score $s_{0, t}^{\text{time}}(x) := \partial_t \log p_t(x)$, define the weighted integrated divergence $\text{BD}_g^\dagger(s_0^{\text{time}}||s_{\theta}^{\text{time}}) :=$

$$\int_0^1 \mathbb{E}_{X_t \sim p_t} \left[\lambda(t) D_g(s_{0, t}^{\text{time}}(X_t) || s_{\theta, t}^{\text{time}}(X_t)) \right] dt,$$

where $s_{\theta, t}^{\text{time}}$ is a differentiable model and $\lambda(t) \geq 0$ is a weighting function. If the model is well specified, minimizing $\text{BD}_g^\dagger(s_0^{\text{time}}||s_{\theta}^{\text{time}})$ recovers the true time score.

Theorem 6.1 (Bregman time-score matching via integration by parts). *Assume $g'(s_{\theta, t}^{\text{time}}(x))$ is differentiable in t for each x . Up to an additive constant independent of θ , $\text{BD}_g^\dagger(s_0^{\text{time}}||s_{\theta}^{\text{time}})$ equals $\text{BD}_g(s_{\theta}^{\text{time}}) :=$*

$$\begin{aligned} & \int_0^1 \mathbb{E}_{X_t \sim p_t} \left[\lambda(t) (g'(s_{\theta, t}^{\text{time}}(X_t)) s_{\theta, t}^{\text{time}}(X_t) - g(s_{\theta, t}^{\text{time}}(X_t))) \right. \\ & \quad \left. + \lambda(t) \partial_t g'(s_{\theta, t}^{\text{time}}(X_t)) + \lambda'(t) g'(s_{\theta, t}^{\text{time}}(X_t)) \right] dt \\ & \quad + \lambda(0) \mathbb{E}_{X_0 \sim q} [g'(s_{\theta, 0}^{\text{time}}(X_0))] - \lambda(1) \mathbb{E}_{X_1 \sim p} [g'(s_{\theta, 1}^{\text{time}}(X_1))]. \end{aligned}$$

The proof is given in Appendix H. For $g(u) = u^2/2$, $\text{BD}_g(s_\theta^{\text{time}})$ reduces to the squared-loss objective used in Time-ScoreMatchingRiesz.

6.2. Automatic Covariate Balancing

Let $r_0(x) := q(x)/p(x)$ denote the endpoint ratio. Time-ScoreMatchingRiesz constructs a log-ratio model by integrating the time score and applying an exponential link: $\ell_\theta(x) := -\int_0^1 s_{\theta,t}^{\text{time}}(x)dt$ and $r_\theta(x) := \exp(\ell_\theta(x))$. Since r_0 satisfies the covariate-balancing identity $\mathbb{E}_{X \sim p}[r_0(X)\varphi(X)] = \mathbb{E}_{X \sim q}[\varphi(X)]$ for all $\varphi \in L_2(p)$, ratio-based representers in ScoreMatchingRiesz can be read as balancing weights.

Loss, link, and model choice. Under matched choices of a Bregman loss and a link, generalized Riesz regression yields approximate sample covariate balancing as a first-order condition over the modeled feature span (Kato, 2026d). In our setting, the generator g in $\text{BD}_g(s_\theta^{\text{time}})$ and the score-to-ratio link ℓ_θ determine the fitting geometry and the implied balancing equations. This viewpoint suggests selecting the score model class and g so that the induced ratio r_θ balances features that are also useful for approximating the outcome regression.

Neyman error minimization. For the orthogonal estimator $\hat{\theta}$, the leading remainder is the error term $\mathbb{E}[(\hat{\alpha} - \alpha_0)(\hat{\gamma} - \gamma_0)]$. The balancing statement should be read as a statement over the chosen feature span. For example, if $\Phi = \text{span}\{\phi_1, \dots, \phi_p\}$ and $\hat{\alpha}$ satisfies

$$\max_{1 \leq j \leq p} \left| \frac{1}{n} \sum_{i=1}^n (\hat{\alpha}(X_i)\phi_j(X_i) - m(W_i, \phi_j)) \right| \leq \lambda,$$

then the empirical Riesz equation is controlled on Φ . If $\hat{\gamma} - \gamma_0$ is well approximated in this span, the same imbalance bound controls the corresponding component of the Neyman remainder. Appendix O records the corresponding balancing identities and a concise sample-level statement.

7. Policy Path Estimation

We introduce a *policy path*—a continuum of policy effects indexed by a scalar policy intensity δ . The path provides a continuous representation of finite-shift policy effects (APEs). Moreover, *only under a known pushforward structure* does the $\delta \rightarrow 0$ limit recover the AME.

7.1. Counterfactual Family and Pushforward

Let $\{P_\delta\}_{\delta \in [-\delta_{\max}, \delta_{\max}]}$ denote the counterfactual laws of the regressor $X \in \mathcal{X}$, with densities $\{p_\delta\}$. A policy is called a *known pushforward* if there exists a *known* measurable map $T_\delta : \mathcal{X} \rightarrow \mathcal{X}$ such that

$$X_\delta = T_\delta(X) \quad \text{for } X \sim P_0,$$

equivalently $P_\delta = (T_\delta)_\# P_0$, where $(T_\delta)_\# P_0$ denotes the pushforward distribution (formal definition and density formulas are given in Appendix G).

7.2. Policy Path as a Continuum of APEs

Define the (symmetric) policy-path parameter

$$\theta_0(\delta) := \mathbb{E}_{X \sim p_{+\delta}}[\gamma_0(X)] - \mathbb{E}_{X \sim p_{-\delta}}[\gamma_0(X)], \quad \delta \in [0, \delta_{\max}],$$

where $\gamma_0(x) := \mathbb{E}[Y \mid X = x]$. For any target shift δ^* , the APE at that shift is simply the point on the path: $\theta_0^{\text{APE}}(\delta^*) = \theta_0(\delta^*)$. Thus, $\delta \mapsto \theta_0(\delta)$ is a *continuous* representation of finite policy effects.

When $p_{\pm\delta} \ll p_0$, we can rewrite $\theta_0(\delta)$ under P_0 using density ratios $r_{\pm\delta}(x) := p_{\pm\delta}(x)/p_0(x)$:

$$\theta_0(\delta) = \mathbb{E}[\alpha_{0,\delta}(X) \gamma_0(X)], \quad \alpha_{0,\delta}(x) := r_{+\delta}(x) - r_{-\delta}(x).$$

Hence, estimating the path reduces to estimating the *path representer* $\alpha_{0,\delta}$, i.e., (a difference of) density ratios.

7.3. The $\delta \rightarrow 0$ Limit

Assume $\delta \mapsto p_\delta(x)$ is differentiable at $\delta = 0$ for a.e. x and define the policy (time) score at the baseline

$$s_0^{\text{pol}}(x) := \partial_\delta \log p_\delta(x)|_{\delta=0}.$$

Then a standard differentiation argument yields

$$\theta'_0(0) = 2 \mathbb{E}[\gamma_0(X) s_0^{\text{pol}}(X)].$$

Crucially, s_0^{pol} depends on *how the policy perturbs the law of X* and is *not* determined by the observational data score

$$s_0^{\text{data}}(x) = \nabla_x \log p_0(x)$$

in general. Therefore, without additional structure, the infinitesimal limit $\theta_0(\delta)/(2\delta)$ is not tied to the AME.

AME limit for shift policies under known pushforward.

Suppose the policy is a *known pushforward* $P_\delta = (T_\delta)_\# P_0$ with $T_0 = \text{Id}$. Then, $\theta_0(\delta) = \mathbb{E}[\gamma_0(T_{+\delta}(X)) - \gamma_0(T_{-\delta}(X))]$ holds, so the local behavior of the path is governed by the geometry of T_δ . In the *translation (shift) policy* $X = (D, Z)$, $T_\delta(d, z) = (d + \delta, z)$, we have

$$\theta_0(\delta) = \mathbb{E}[\gamma_0(D + \delta, Z) - \gamma_0(D - \delta, Z)]$$

and, under mild smoothness, we have

$$\lim_{\delta \downarrow 0} \frac{\theta_0(\delta)}{2\delta} = \mathbb{E}[\partial_d \gamma_0(D, Z)] =: \theta_0^{\text{AME}}.$$

Thus, the policy path recovers AME as its $\delta \rightarrow 0$ limit *only in the structured (known-pushforward) regime*. Appendix G further shows that the same structure allows construction of $\alpha_{0,\delta}$ (hence APEs and the full path) from the data score.

Coincidence of the Data Score and Time Score A one-parameter policy family $\{p_\delta\}$ itself provides a natural “time” index. Define the time score along the policy as $s_\delta^{\text{time}}(x) := \partial_\delta \log p_\delta(x)$. Then s_δ^{time} is exactly the policy score. For translation pushforwards,

$$s_\delta^{\text{time}}(d, z) = \partial_\delta \log p_\delta(d, z) = -\partial_d \log p_0(d - \delta, z),$$

so the time score along the policy *coincides* with the AME representer evaluated along the known path. This is the basic reason why, under known pushforwards, a single treatment-direction data-score model on P_0 can generate ratios for all δ via one-dimensional integration (Appendix G).

8. Theoretical Analysis

This section provides theoretical analysis related to our proposed ScoreMatchingRiesz. Because our nuisance components are constructed from either the *data score* $\nabla_x \log p(x)$ or the *time score* $\partial_t \log p_t(x)$, we focus on L_2 -type guarantees for these scores.

8.1. Data-ScoreMatchingRiesz

This subsection records L_2 -type score estimation rates from diffusion-model-based (denoising) score matching, as established by [Oko et al. \(2023\)](#). Throughout, let p_0 be a probability density on a bounded domain $\mathcal{X} \subset \mathbb{R}^d$ (e.g., $\mathcal{X} = [-1, 1]^d$), and let $X \sim p_0$. Recall that for $\sigma > 0$, we defined the smoothed density $p_\sigma := p_0 * \mathcal{N}(0, \sigma^2 I_d)$ (so $\tilde{X}_\sigma \sim p_\sigma$), and its (data) score $\nabla_x \log p_\sigma(x)$ for $x \in \mathcal{X}$. Let $\hat{s}_\sigma^{\text{data}}(x)$ denote an estimator of this score.

Additional notations and assumptions. For a function $g: \mathcal{X} \rightarrow \mathbb{R}^d$ and a density p on $\mathcal{X}$, define $\|g\|_{L_2(p)}^2 := \mathbb{E}_{X \sim p}[\|g(X)\|_2^2]$. For a score field $g(\cdot, \sigma)$ depending on $\sigma \in [\sigma_{\min}, \sigma_{\max}]$, define the integrated L_2 norm $\|g\|_\nu^2 := \int_{\sigma_{\min}}^{\sigma_{\max}} \|g(\cdot, \sigma)\|_{L_2(p_\sigma)}^2 d\sigma$, where $0 < \sigma_{\min} < \sigma_{\max} < \infty$. We write $a_n \lesssim b_n$ if $a_n \leq C b_n$ for a universal constant $C > 0$ and all large n . We write $\text{polylog}(n)$ for a factor bounded by $(\log n)^c$ for some $c > 0$. Let $B_{p,q}^s(\mathcal{X})$ denote a Besov space on $\mathcal{X}$ with smoothness $s > 0$ and integrability parameters $p, q \in [1, \infty]$. We assume $p_0 \in B_{p,q}^s(\mathcal{X})$.

L_2 rate. [Oko et al. \(2023\)](#) shows the following result, which is directly applicable to Data-ScoreMatchingRiesz.

Theorem 8.1 (L_2 rate for Data-ScoreMatchingRiesz; [Oko et al. \(2023\)](#)). *Under the assumptions of [Oko et al. \(2023\)](#), the Data-ScoreMatchingRiesz estimator $\hat{s}^{\text{data}}$ can be constructed so that*

$$\mathbb{E} \left[\|\hat{s}_\sigma^{\text{data}} - \nabla \log p_\sigma\|_\nu^2 \right] \lesssim n^{-2s/(2s+d)} \cdot \text{polylog}(n).$$

Moreover, the minimax lower bound over the same Besov class matches $n^{-2s/(2s+d)}$ up to logarithmic factors.

In AME estimation with $X = (D, Z)$ and continuous D , the Riesz representer is $\alpha_0^{\text{AME}}(D, Z) = -\partial_d \log p_0(D, Z)$ (equivalently, $-\{s_0^{\text{data}}(D, Z)\}_d$).

8.2. Time-ScoreMatchingRiesz

We next give an L_2 rate for Time-ScoreMatchingRiesz. Since the formulation varies across applications, we consider the following representative problem.

Additional notations and assumptions. Let q and p be two probability densities on $\mathcal{X}$ such that $q \ll p$. Define the density ratio as $r(x) := \frac{q(x)}{p(x)}$ for $x \in \mathcal{X}$. Let $\{p_t\}_{t \in [0,1]}$ be a family of *bridge* densities connecting q and p : $p_0 = q$ and $p_1 = p$. We assume $t \mapsto p_t(x)$ is differentiable for a.e. x , and define the *time score* $s_t^{\text{time}}(x) := \partial_t \log p_t(x)$ for $(x, t) \in \mathcal{X} \times (0, 1)$. Write $X_t \sim p_t$.

Let $\lambda: [0, 1] \rightarrow \mathbb{R}_+$ be a measurable weighting function and define $\lambda_{\min} := \inf_{t \in [0,1]} \lambda(t)$. When $\lambda_{\min} > 0$, define the weighted L_2 norm for a family $\{f_t\}_{t \in [0,1]}$ with $f_t: \mathcal{X} \rightarrow \mathbb{R}$ by $\|f\|_{\mu_\lambda}^2 := \int_0^1 \mathbb{E}_{X_t \sim p_t}[\lambda(t) f_t(X_t)^2] dt$. We write μ_1 for the special case $\lambda(t) \equiv 1$.

Let P_n denote an effective complexity parameter for $\mathcal{F}_n$ (e.g., number of parameters or an upper bound thereof), used to express approximation and generalization rates.

Assumption 8.1 (Overlap of p_t). *There exist constants $0 < c \leq C < \infty$ such that for all $t \in [0, 1]$ and for a.e. $x \in \mathcal{X}$, $cp(x) \leq p_t(x) \leq Cp(x)$ holds.*

Assumption 8.1 is a technical device to transfer L_2 control along the bridge to downstream bounds for ratios and representers. Appendix M.3 summarizes practical bridge and weighting choices that mitigate violations of overlap.

Assumption 8.2 (Smoothness and approximation rate). *The map $(x, t) \mapsto s_t^{\text{time}}(x)$ belongs to a Besov ball on $\mathcal{X} \times [0, 1]$ with smoothness $s > 0$ (e.g. $B_{2,2}^s(\mathcal{X} \times [0, 1])$), and the model class $\mathcal{F}_n$ satisfies: there exists $s_n^* \in \mathcal{F}_n$ such that $\|s_n^* - s^{\text{time}}\|_{\mu_\lambda}^2 \lesssim P_n^{-2s/(d+1)}$.*

Assumption 8.3 (Uniform deviation and complexity). *There exists a (data-dependent) random variable Δ_n such that with high probability, $\sup_{s \in \mathcal{F}_n} |\hat{\mathcal{R}}_n(s) - \mathcal{R}(s)| \leq \Delta_n$ holds, where $\Delta_n \lesssim \frac{P_n \text{polylog}(n)}{n}$.*

L_2 rate. Let $\mathcal{R}(\cdot)$ denote the population time-score-matching objective in Eq. (2). Under regularity conditions (as in Lemma J.1), s_t^{time} is the population minimizer of $\mathcal{R}$. Let $\hat{\mathcal{R}}_n(\cdot)$ be an empirical counterpart of $\mathcal{R}(\cdot)$ computed from samples from q and p and Monte Carlo bridge draws as in Section 3. Let $\mathcal{F}_n$ be a class of time-score models $s_{\theta,t}^{\text{time}}(x)$ (e.g., neural networks) that are differentiable in t . Define the empirical risk minimizer as $\hat{s} \in \arg \min_{s \in \mathcal{F}_n} \hat{\mathcal{R}}_n(s)$. Then, the following result holds.

Theorem 8.2 (L_2 rate for Time-ScoreMatchingRiesz).

Suppose Assumptions 8.1–8.3 hold and $\lambda_{\min} > 0$. Then,

$$\|\hat{s} - s^{\text{time}}\|_{\mu_\lambda}^2 \lesssim \inf_{s \in \mathcal{F}_n} \|s - s^{\text{time}}\|_{\mu_\lambda}^2 + \Delta_n$$

holds. In particular, under Assumptions 8.2 and 8.3,

$$\|\hat{s} - s^{\text{time}}\|_{\mu_\lambda}^2 \lesssim P_n^{-2s/(d+1)} + \frac{P_n \text{polylog}(n)}{n}$$

holds. Choosing $P_n \asymp n^{(d+1)/(2s+d+1)}$ yields the (near-)optimal rate

$$\|\hat{s} - s^{\text{time}}\|_{\mu_\lambda} = O_{\mathbb{P}}\left(n^{-s/(2s+d+1)} \cdot \text{polylog}(n)\right).$$

The proof is given in Appendix J.

Dimension dependence. The exponent contains $d + 1$ because Time-ScoreMatchingRiesz estimates a time score on $\mathcal{X} \times [0, 1]$. The extra dimension is the bridge index and not an additional covariate block. The result does not remove the usual nonparametric dependence on d ; in high-dimensional applications, the usefulness of the bound relies on structural approximation by the chosen score model class and on regularization.

Integrating an estimated time score $\hat{s}$ yields a plug-in estimator of $\log r$ and hence of r . Appendix K records the corresponding L_2 rate for the Riesz representer.

8.3. Efficient Estimation of the Parameter of Interest

We recall the asymptotic efficiency result for

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n \left(m(W_i, \hat{\gamma}) + \hat{\alpha}(X_i)(Y_i - \hat{\gamma}(X_i)) \right).$$

From the basic results in semiparametric analysis, as shown in van der Vaart (1998) and Chernozhukov et al. (2018), the following result holds.

Theorem 8.3 (Asymptotic normality). Assume that both $\hat{\gamma}$ and $\hat{\alpha}$ satisfy the Donsker condition or are constructed via cross-fitting. Assume that the following conditions hold:

$$\begin{aligned} \|\hat{\alpha} - \alpha_0\|_2 &= o_p(1), \quad \|\hat{\gamma} - \gamma_0\|_2 = o_p(1), \\ \|\hat{\alpha} - \alpha_0\|_2 \|\hat{\gamma} - \gamma_0\|_2 &= o_p(1/\sqrt{n}). \end{aligned}$$

Then, it holds that

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \mathbb{E}[\psi(W; \eta_0, \theta_0)^2]) \quad (n \rightarrow \infty).$$

By connecting the error analysis of ScoreMatchingRiesz with the asymptotic normality result, we obtain the following corollary.

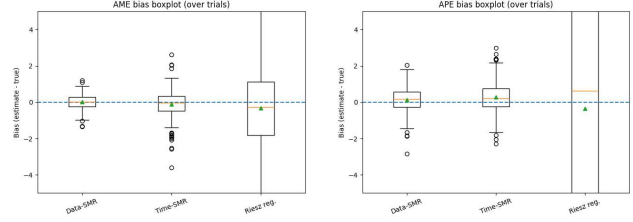
(a) AME (true = 2.225).

Metric	Data-SMR	Time-SMR	Riesz reg.
Bias	0.001	-0.075	-0.205
MSE	0.161	0.637	10.076
Cov.	91.8%	90.9%	86.9%

(b) APE (true = 4.063).

Metric	Data-SMR	Time-SMR	Riesz reg.
Bias	0.131	0.268	-0.152
MSE	0.486	1.117	74.918
Cov.	89.4%	75.0%	85.2%

Table 2. Performance metrics for AME and APE.



(a) Results in AME estimation (b) Results in APE estimation

Figure 1. Box plots of errors (biases) of the estimates

Corollary 8.4 (L_2 rate for Riesz representers). In AME estimation, $\alpha_0^{\text{AME}}(D, Z) = -\partial_d \log p_0(D, Z)$. If Data-ScoreMatchingRiesz is evaluated at a small noise level σ_{eval} , then $\hat{\alpha}^{\text{AME}}(d, z) = -(\hat{s}_{\sigma_{\text{eval}}}(d, z))_d$, and

$$\begin{aligned} \|\hat{\alpha}^{\text{AME}} - \alpha_0^{\text{AME}}\|_{L_2(P_0)} &\leq \|(\hat{s}_{\sigma_{\text{eval}}} - s_{\sigma_{\text{eval}}})_d\|_{L_2(P_0)} + \|(s_{\sigma_{\text{eval}}} - s_0)_d\|_{L_2(P_0)}. \end{aligned}$$

The first term is the score-estimation error, and the second term is the smoothing bias discussed in Appendix L.

For ratio-based targets, including ATE, APE, and each fixed- δ policy-path point, construct $\widehat{\log r}$ by integrating an estimated score field and apply clipping before exponentiation. Assume that $|\log r| \leq c$ almost surely and clip $\widehat{\log r}$ to $[-c, c]$. Then, we have

$$\begin{aligned} \|\widehat{\log r} - \log r\|_{L_2(p)} &\lesssim \|\hat{s}^{\text{time}} - s^{\text{time}}\|_{\mu_1}, \\ \|\hat{r} - r\|_{L_2(p)} &\lesssim e^c \|\widehat{\log r} - \log r\|_{L_2(p)}, \\ \|\hat{\alpha} - \alpha_0\|_2 &\lesssim \|\hat{r}_+ - r_+\|_2 + \|\hat{r}_- - r_-\|_2 \end{aligned}$$

for the corresponding ratio-based representer α_0 . Consequently, if $\|\hat{\gamma} - \gamma_0\|_2 = O_{\mathbb{P}}(n^{-b})$, the preceding Time-ScoreMatchingRiesz rate satisfies the product condition in Theorem 8.3 whenever $\frac{s}{2s+d+1} + b > \frac{1}{2}$. For Data-ScoreMatchingRiesz, the smoothing-bias term in the first display must also satisfy the corresponding product condition.

9. Experiments

We compare Data-ScoreMatchingRiesz (Data-SMR) and Time-ScoreMatchingRiesz (Time-SMR) with Riesz regression using the synthetic and semi-synthetic datasets. The replication codes are available as example codes in the `genriesz` library (Kato, 2026c)¹

¹<https://github.com/MasaKat0/genriesz/tree/main/notebooks/scorematchingriesz>

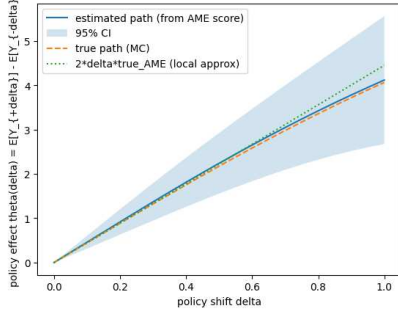


Figure 2. Policy path visualization

All methods use the same MLP architecture for $\hat{\gamma}$ and two-fold cross-fitting. Additional results are reported in Appendices P and Q. Appendix P includes two 50-dimensional pushforward designs and a non-pushforward stochastic-intervention design. In the non-pushforward design, the observational data score does not determine the APE or policy-path density ratios, so Data-SMR is unavailable and Time-SMR is the applicable score-based construction; joint Time-SMR reduces the APE MSE from 7.392 to 1.254 in that design.

9.1. Simulation Studies

We generate $X = (X^{(1)}, X^{(2)}, X^{(3)}) \in \mathbb{R}^3$ from $X \sim \mathcal{N}(0, \Sigma)$ with $\Sigma = \begin{pmatrix} 1 & 0.1 & 0.1 \\ 0.1 & 1 & 0.1 \\ 0.1 & 0.1 & 1 \end{pmatrix}$. We set $D := X^{(1)}$ and $Z := (X^{(2)}, X^{(3)})$. The outcome is $Y = \mu(X) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 1)$ and $\mu(x) = 1 + x^{(1)} + 0.1(x^{(1)})^2 + 2 \sin(x^{(1)}) + x^{(2)} + x^{(1)}x^{(2)} + (x^{(3)})^2 + (x^{(3)})^3$.

Targets. We estimate the AME $\theta_0^{\text{AME}} = \mathbb{E}[\partial_d \gamma_0(D, Z)]$ and the APE under the shift policy at $\delta = 1$, $\theta_0^{\text{APE}} = \mathbb{E}[\mu(D + 1, Z)] - \mathbb{E}[\mu(D - 1, Z)]$. The ground truths are computed by population Monte Carlo.

Evaluation metrics. Across 200 trials, we report (i) bias and MSE of $\hat{\theta}$ and (ii) empirical coverage of 95% confidence intervals. For the shift policy, we also estimate the full policy path $\delta \mapsto \theta_0(\delta)$ on a grid (Section 7) and plot pointwise 95% bands, the true path, and the local linear approximation $2\delta \theta_0^{\text{AME}}$.

Results. For simulation studies, Tables 2a and 2b show that Data-SMR has the smallest MSE in this baseline non-linear design: 0.161 for AME and 0.486 for APE at $\delta = 1$. The empirical coverage is below the nominal 95% level for all methods, especially for Time-SMR in APE, where the coverage is 75.0%. Riesz regression is less stable in this design, with MSE 10.076 for AME and 74.918 for APE. Figure 2 shows that the policy path departs from the local linear approximation $2\delta \theta_0^{\text{AME}}$ as δ increases, which motivates estimating the full path rather than relying only on the local AME approximation. Additional experiments in Appendix P include high-dimensional pushforward de-

Table 3. IHDP semi-synthetic ATE benchmark.

Method	Bias	RMSE	95% cov.
Time-SMR	0.937	1.571	100.0%
Joint-SMR	0.860	1.478	100.0%
SQ-Riesz	19.111	27.954	0.0%
BKL-Riesz	48.294	84.072	19.0%
MLE	11.548	19.643	46.0%

signs and a non-pushforward stochastic-intervention design in which the data-score construction is unavailable. Under the product-rate condition in Theorem 8.3, asymptotic coverage is expected once second-order nuisance errors vanish. In these finite-sample experiments, however, the confidence intervals should be interpreted with caution. The undercoverage is consistent with residual second-order nuisance error, ratio amplification after exponentiation, weak overlap near bridge endpoints, and smoothing bias in Data-SMR.

9.2. IHDP Semi-Synthetic ATE Benchmark

We also evaluate the ATE setting on the IHDP semi-synthetic benchmark, which uses real covariates and known ground-truth potential-outcome means. We use the observational training split with 100 replications, each with $n = 672$ units and 25 covariates. For this binary-treatment benchmark, the Riesz representer is ratio-based, so we compare Time-SMR and its joint-training variant, denoted Joint-SMR, with direct ratio-fitting baselines and AIPW with a logistic propensity score. Table 3 reports bias, RMSE, and empirical 95% coverage. Joint-SMR has the smallest RMSE, followed by Time-SMR. Both score-based estimators have substantially smaller RMSE than the compared baselines, while their confidence intervals are conservative in this benchmark. We interpret these results as supplementary evidence on real covariates and additional baselines, rather than as evidence of uniform dominance. The full setup and additional metrics are reported in Appendix R.

10. Conclusion

We proposed ScoreMatchingRiesz to estimate the Riesz representer using data scores and time scores, which provides a unified interpretation across causal targets. Data-ScoreMatchingRiesz is effective when the representer is recoverable from the observational score, such as AME and APEs, whereas Time-ScoreMatchingRiesz is needed when endpoint ratios must be learned through a bridge. Time-ScoreMatchingRiesz is computationally heavier for a single target because it requires bridge sampling, time derivatives, and score integration, but a trained score model can be reused across multiple APEs and policy-path points. The policy path organizes finite-shift APEs and recovers AME under known-pushforward structure, making score matching a useful Riesz representer estimation module for causal inference.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

References

- Susan Athey and Stefan Wager. Policy learning with observational data. *Econometrica*, 89(1):133–161, 2021.
- Susan Athey, Guido W. Imbens, and Stefan Wager. Approximate residual balancing: debiased inference of average treatment effects in high dimensions. *Journal of the Royal Statistical Society. Series B (Statistical Methodology)*, 80(4):597–623, 2018.
- Heejeung Bang and James M. Robins. Doubly robust estimation in missing data and causal inference models. *Biometrics*, 61(4):962–973, 2005.
- Eli Ben-Michael, Avi Feller, David A. Hirshberg, and José R. Zubizarreta. The balancing act in causal inference, 2021. arXiv: 2110.14831.
- P.J. Bickel, C.A.J. Klaassen, Y. Ritov, and J.A. Wellner. *Efficient and Adaptive Estimation for Semiparametric Models*. Johns Hopkins series in the mathematical sciences. Springer New York, 1998.
- Steffen Bickel, Michael Brückner, and Tobias Scheffer. Discriminative learning under covariate shift. *J. Mach. Learn. Res.*, pp. 2137–2155, December 2009.
- David Bruns-Smith, Oliver Dukes, Avi Feller, and Elizabeth L Ogburn. Augmented balancing weights as linear regression. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 04 2025.
- Xiaohong Chen and Zhipeng Liao. Sieve semiparametric two-step gmm under weak dependence. *Journal of Econometrics*, 189(1):163–186, 2015.
- Xiaohong Chen and Demian Pouzo. Sieve wald and qlr inferences on semi/nonparametric conditional moment models. *Econometrica*, 83(3):1013–1079, 2015.
- Xiaohong Chen, Zhipeng Liao, and Yixiao Sun. Sieve inference on possibly misspecified semi-nonparametric time series models. *Journal of Econometrics*, 178:639–658, 2014.
- Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 2018.
- Victor Chernozhukov, Whitney K. Newey, Victor Quintas-Martínez, and Vasilis Syrgkanis. Automatic debiased machine learning via riesz regression, 2021. arXiv:2104.14737.
- Victor Chernozhukov, Whitney Newey, Víctor M Quintas-Martínez, and Vasilis Syrgkanis. RieszNet and ForestRiesz: Automatic debiased machine learning with neural nets and random forests. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 3901–3914. PMLR, 2022a.
- Victor Chernozhukov, Whitney K. Newey, and Rahul Singh. Automatic debiased machine learning of causal and structural effects. *Econometrica*, 90(3):967–1027, 2022b.
- Victor Chernozhukov, Christian Hansen, Nathan Kallus, Martin Spindler, and Vasilis Syrgkanis. *Applied Causal Inference Powered by ML and AI*. CausalML-book.org, 2024a. arXiv:2403.02467.
- Victor Chernozhukov, Whitney Newey, Rahul Singh, and Vasilis Syrgkanis. Adversarial estimation of riesz representers, 2024b. arXiv: 2101.00009.
- Victor Chernozhukov, Sokbae Lee, Adam M. Rosen, and Liyang Sun. Policy learning with confidence, 2025a. arXiv: 2502.10653.
- Victor Chernozhukov, Michael Newey, Whitney K Newey, Rahul Singh, and Vasilis Srygkanis. Automatic debiased machine learning for covariate shifts, 2025b. arXiv: 2307.04527.
- Kristy Choi, Chenlin Meng, Yang Song, and Stefano Ermon. Density ratio estimation via infinitesimal classification. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2022.
- Michael U. Gutmann and Jun-ichiro Hirayama. Bregman divergence as general framework to estimate unnormalized statistical models. In *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2011.
- Jens Hainmueller. Entropy balancing for causal effects: A multivariate reweighting method to produce balanced samples in observational studies. *Political Analysis*, 20(1):25–46, 2012.
- Chad Hazlett. Kernel balancing: a flexible non-parametric weighting procedure for estimating causal effects. *Statistica Sinica*, 30:1155–1189, 2020.
- Keisuke Hirano, Guido Imbens, and Geert Ridder. Efficient estimation of average treatment effects using the estimated propensity score. *Econometrica*, 2003.

- Aapo Hyvärinen. Estimation of non-normalized statistical models by score matching. *Journal of Machine Learning Research*, 6(24):695–709, 2005.
- I. A. Ibragimov and R. Z. Khas’minskii. *Statistical estimation : asymptotic theory*. Applications of mathematics. Springer-Verlag, 1981.
- Òscar Jordà. Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1):161–182, 2005.
- Takafumi Kanamori, Shohei Hido, and Masashi Sugiyama. A least-squares approach to direct importance estimation. *Journal of Machine Learning Research*, 10(Jul.):1391–1445, 2009.
- Masahiro Kato. Direct bias-correction term estimation for propensity scores and average treatment effect estimation, 2025a. arXiv: 2509.22122.
- Masahiro Kato. Direct bias-correction term estimation for average treatment effect estimation, 2025b. openreview.
- Masahiro Kato. Nearest neighbor matching as least squares density ratio estimation and riesz regression, 2025c. arXiv: 2510.24433.
- Masahiro Kato. Direct bias-correction term estimation for average treatment effect estimation, 2026a. arXiv: 2509.22122v2.
- Masahiro Kato. Covariate balancing and riesz regression should be guided by the neyman orthogonal score in debiased machine learning, 2026b. arXiv: 2605.06386.
- Masahiro Kato. genriesz: A python package for automatic debiased machine learning with generalized riesz regression, 2026c. arXiv: 2602.17543.
- Masahiro Kato. A unified framework for debiased machine learning: Riesz representer fitting under bregman divergence, 2026d. arXiv: 2601.07752.
- Masahiro Kato. Scorematchingriesz for automatic debiased machine learning and policy path estimation with an application to japanese monetary policy evaluation. *JSAI Technical Report, SIGFIN*, 2026(FIN-036), 2026e.
- Masahiro Kato and Takeshi Teshima. Non-negative bregman divergence minimization for deep direct density ratio estimation. In *International Conference on Machine Learning (ICML)*, 2021.
- Masahiro Kato, Kota Matsui, and Ryo Inokuchi. Double debiased covariate shift adaptation robust to density-ratio estimation, 2024. arXiv: 2310.16638.
- Chris A. J. Klaassen. Consistent estimation of the influence function of locally asymptotically linear estimators. *Annals of Statistics*, 15, 1987.
- B. Ya. Levit. On the efficiency of a class of non-parametric estimates. *Theory of Probability & Its Applications*, 20(4):723–740, 1976.
- Zhexiao Lin, Peng Ding, and Fang Han. Estimation based on nearest neighbor matching: from density ratio to average treatment effect. *Econometrica*, 91(6):2187–2217, 2023.
- Siwei Lyu. Interpretation and generalization of score matching. In *Conference on Uncertainty in Artificial Intelligence (UAI)*. AUAI Press, 2009. ISBN 9780974903958.
- Whitney K. Newey. The asymptotic variance of semiparametric estimators. *Econometrica*, 62(6), 1994.
- Kazusato Oko, Shunta Akiyama, and Taiji Suzuki. Diffusion models are minimax optimal distribution estimators. In *International Conference on Machine Learning (ICML)*, 2023.
- J. Pfanzagl and W. Wefelmeyer. *Contributions to a General Asymptotic Statistical Theory*. Lecture notes in statistics. Springer-Verlag, 1982.
- B. Rhodes, K. Xu, and M.U. Gutmann. Telescoping density-ratio estimation. In *NeurIPS*, 2020.
- J. M. Robins, A. Rotnitzky, and L. P. Zhao. Estimation of regression coefficients when some regressors are not always observed. *Journal of the American Statistical Association*, 89:846–866, 1994.
- Alejandro Schuler, Michael Baiocchi, Robert Tibshirani, and Nigam Shah. A comparison of methods for model selection when estimating individual treatment effects, 2018.
- Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *International Conference on Learning Representations (ICLR)*, 2021.
- Yang Song and Stefano Ermon. Generative modeling by estimating gradients of the data distribution, 2020. arXiv: 1907.05600.
- Masashi Sugiyama, Taiji Suzuki, and Takafumi Kanamori. Density ratio matching under the bregman divergence: A unified framework of density ratio estimation. *Annals of the Institute of Statistical Mathematics*, 64, 10 2011.
- Masashi Sugiyama, Taiji Suzuki, and Takafumi Kanamori. *Density Ratio Estimation in Machine Learning*. Cambridge University Press, 2012.

- Masatoshi Uehara, Masahiro Kato, and Shota Yasui. Off-policy evaluation and learning for external validity under a covariate shift. In *Conference on Neural Information Processing Systems (NeurIPS)*, 2020.
- van der Laan. Targeted maximum likelihood learning, 2006. U.C. Berkeley Division of Biostatistics Working Paper Series. Working Paper 213.
- Aad W. van der Vaart. *Asymptotic Statistics*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 1998.
- Pascal Vincent. A connection between score matching and denoising autoencoders. *Neural Computation*, 23(7):1661–1674, 2011.
- Qingyuan Zhao. Covariate balancing propensity score by tailored loss functions. *The Annals of Statistics*, 47(2): 965 – 993, 2019.
- Wenjing Zheng and Mark J. van der Laan. *Cross-Validated Targeted Minimum-Loss-Based Estimation*, pp. 459–474. Springer New York, 2011.
- José R. Zubizarreta. Stable weights that balance covariates for estimation with incomplete outcome data. *Journal of the American Statistical Association*, 110(511):910–922, 2015.

A. Related Work

This appendix complements the brief discussion in the main text and places ScoreMatchingRiesz in the intersection of (i) semiparametric efficiency and DML, (ii) Riesz-representer estimation and covariate balancing, and (iii) score estimation and density-ratio estimation via score matching and infinitesimal classification.

A.1. Efficient estimation, orthogonal scores, and DML

A recurring theme in semiparametric statistics is that (asymptotically) efficient estimation can be achieved by augmenting a plug-in estimator with a correction term based on an influence function. Classical constructions include one-step estimators and estimating-equation (Z-estimation) approaches (Levit, 1976; Ibragimov & Khas'minskii, 1981; Pfanzagl & Wefelmeyer, 1982; Bickel et al., 1998; van der Vaart, 1998; Newey, 1994). Related ideas also underpin sieve-based inference for (possibly irregular) functionals, where Riesz representers play an explicit role in characterizing local sensitivity and efficient variance (Chen et al., 2014; Chen & Liao, 2015; Chen & Pouzo, 2015).

In missing-data and causal-inference settings, these influence-function constructions yield doubly robust and augmented inverse-probability-weighted scores (Robins et al., 1994; Bang & Robins, 2005; Hirano et al., 2003) and targeted learning estimators that explicitly target the efficient influence function (van der Laan, 2006; Schuler et al., 2018). Debiased (or “double”) machine learning extends these principles to high-dimensional and nonparametric nuisance estimation by leveraging Neyman orthogonality and cross-fitting to control the effect of first-stage regularization (Chernozhukov et al., 2018; Klaassen, 1987; Zheng & van der Laan, 2011). The *automatic* debiased ML formulation emphasizes that for general linear and continuous functionals of the regression function, efficient influence functions involve a *Riesz representer* (or, in causal examples, a set of balancing weights) that must itself be estimated (Chernozhukov et al., 2022b). Our paper follows this line and focuses on the representer-estimation subproblem, proposing score-matching-based modules that can be plugged into Neyman orthogonal scores.

A.2. Riesz representers, Riesz regression, and covariate balancing

Riesz regression and its variants. A direct approach to representer estimation is *Riesz regression*, which fits α by minimizing a sample analog of the Hilbert-space quadratic characterization of the Riesz representer (Chernozhukov et al., 2021; 2022b). In AME settings, this objective coincides with Hyvärinen score matching and (in certain linear models) with LSIF-type density-ratio objectives (Hyvärinen, 2005; Kanamori et al., 2009). Alternative representer estimators include matching and nearest-neighbor methods that directly approximate the Riesz functional (Lin et al., 2023; Kato, 2025c). For ATE, the representer reduces to inverse propensity weights, which are also the key ingredient in classical efficient ATE estimators (Hirano et al., 2003; Bang & Robins, 2005).

Balancing weights as ratio-based representers. In many targets, including ATE, ATT, transportability, and covariate shift, the Riesz representer is a density ratio or a simple transform of one, so representer estimation is equivalent to constructing weights that *balance* covariate moments between two distributions. This connection links representer estimation to a large literature on covariate balancing and stable weighting, including entropy balancing (Hainmueller, 2012), stable balancing weights (Zubizarreta, 2015), approximate residual balancing (Athey et al., 2018; Ben-Michael et al., 2021), kernel balancing (Hazlett, 2020), and tailored loss minimization and covariate-balancing perspectives on efficiency (Zhao, 2019). Recent work shows that such weighting schemes can be interpreted through generalized Riesz regression objectives built from Bregman divergences, clarifying the role of the loss and link function in determining which balancing weights are favored (Kato, 2026d; Bruns-Smith et al., 2025). Section 6 adapts this “loss–link” viewpoint to our score-based construction: the time-score loss (Theorem 6.1) and the score-to-ratio link together determine the implied balancing weights, and the resulting (approximate) balancing identities help control the Neyman remainder in orthogonal estimation.

A.3. Score matching and diffusion models

Score matching estimates the score $\nabla_x \log p(x)$ without estimating the density normalizing constant. Hyvärinen’s original formulation minimizes a Fisher-divergence objective that becomes computable via integration by parts (Hyvärinen, 2005). A practical challenge is that the Hyvärinen objective involves a divergence (Jacobian-trace) term, which can be expensive or unstable in high dimensions. Denoising score matching (DSM) addresses this by training on Gaussian-perturbed samples, replacing the target score by the score of a smoothed density (Vincent, 2011). DSM is the workhorse of modern score-based generative modeling and diffusion models, where one trains a single network over a continuum of noise levels

(Song & Ermon, 2020; Song et al., 2021).

While diffusion models are typically used for density estimation and sampling, their statistical theory provides off-the-shelf nonparametric guarantees for score estimation. In particular, [Okó et al. \(2023\)](#) establish minimax-optimal (up to logs) convergence rates for diffusion-model score estimators, which we leverage as representer rates for Data-ScoreMatchingRiesz (Section 8.1). Our use of DSM is therefore conceptually different from generative modeling: we employ score learning as a *Riesz-representer module* inside orthogonal score equations, and we propagate score error to the error of the representer and of the final debiased estimator.

A.4. Density ratio estimation, telescoping bridges, and infinitesimal classification

Density ratio estimation (DRE)—estimating q/p from samples—is a central primitive in covariate shift, importance weighting, and off-policy evaluation ([Sugiyama et al., 2012](#); [Uehara et al., 2020](#)). Representative approaches include least-squares importance fitting (LSIF) ([Kanamori et al., 2009](#)) and its generalizations based on Bregman divergences ([Sugiyama et al., 2011](#)), as well as classifier-based ratio estimation ([Bickel et al., 2009](#)). DRE also underlies debiased estimation under distribution shift, where the relevant representers are ratios between the target and source covariate laws ([Uehara et al., 2020](#); [Kato et al., 2024](#); [Chernozhukov et al., 2025b](#)).

A key practical issue is that highly flexible ratio models can overfit badly, producing extreme weights and unstable downstream estimators. This phenomenon has been documented as “train loss hacking” ([Kato & Teshima, 2021](#)) and as the “density chasm” in high dimensions ([Rhodes et al., 2020](#)). One mitigation strategy is to replace direct endpoint-to-endpoint ratio estimation by a product of intermediate ratios along a bridge ([Rhodes et al., 2020](#)). Taking a continuum limit leads to *infinitesimal classification* and the DRE- ∞ objective, which learns the time score $\partial_t \log p_t(x)$ along the bridge and recovers the log ratio by integration ([Choi et al., 2022](#)). Our Time-ScoreMatchingRiesz adopts this mechanism as a representer estimator when the target representer is ratio-based (ATE/APE/policy paths), and our analysis focuses on the error metrics needed for Neyman orthogonal debiased estimation rather than on ratio estimation alone.

A.5. Policy effects under distributional shifts and policy paths

Counterfactual policy evaluation can often be phrased as estimating a regression functional under a shifted distribution of covariates. In economics, these objects include policy counterfactuals and structural policy effects, and in macro-finance they arise naturally in local projection analyses ([Jordà, 2005](#)). In machine learning and econometrics, related themes appear in policy learning and off-policy evaluation, where the goal is to evaluate or optimize the value of (possibly stochastic) policies ([Athey & Wager, 2021](#); [Chernozhukov et al., 2025a](#); [Uehara et al., 2020](#)). Our “policy path” parameter is a descriptive object that traces policy effects continuously as the intervention intensity varies.

Methodologically, our paper emphasizes that the representer for a finite policy shift is typically a density ratio, and hence a balancing weight. When the counterfactual law is a *known pushforward* of the observational law, density ratios can be expressed as line integrals of the observational data score along the known transport path, enabling reuse of a single score model across many policy shifts ([Kato, 2025a;b](#)). Outside the pushforward regime (e.g. stochastic interventions that resample $D \mid Z$), this score-to-ratio construction is unavailable and one must learn ratios (or time scores) from endpoint samples, motivating our Time-ScoreMatchingRiesz constructions.

B. Derivation of the Hyvärinen Score Matching Objective

Concretely, expanding $\mathcal{R}^\dagger$ yields

$$\mathcal{R}^\dagger(s) = \frac{1}{2} \mathbb{E} [\|s(X)\|^2] - \mathbb{E} [s(X)^\top \nabla_x \log p_0(X)] + \text{const.}$$

Assuming boundary decay so that integration by parts applies, we have

$$\mathbb{E} [s(X)^\top \nabla_x \log p_0(X)] = -\mathbb{E} [\nabla_x \cdot s(X)],$$

where $\nabla_x \cdot s(x) = \text{tr}\{\nabla_x s(x)\}$ is the divergence of the vector field s . Substituting this identity into the expansion yields Eq. (1). Thus, minimizing Eq. (1) over a function class estimates the score without ever estimating p_0 itself.

AME as one-coordinate Hyvärinen score matching. For AME, $m(W, \gamma) = \partial_d \gamma(D, Z)$. The quadratic Riesz regression objective is

$$\mathcal{L}_{\text{RR}}(\alpha) = \mathbb{E} \left[\frac{1}{2} \alpha(D, Z)^2 - \partial_d \alpha(D, Z) \right].$$

Let $u(D, Z) = -\alpha(D, Z)$. Then, up to the sign change defining the AME representer, the objective becomes

$$\mathbb{E} \left[\frac{1}{2} u(D, Z)^2 + \partial_d u(D, Z) \right],$$

which is the Hyvärinen objective restricted to the treatment coordinate.

C. Equivalence of “Time Scores” in DSM and DRE- ∞ Under a Shared Bridge

In DRE- ∞ , the *time score* associated with a differentiable bridge of densities $\{p_t\}_{t \in [0,1]}$ is defined as

$$s^{\text{time}}(x, t) := \partial_t \log p_t(x).$$

This definition is purely path-based: s^{time} depends on the chosen bridge $\{p_t\}$.

A DSM forward noising mechanism also induces a one-parameter family of marginal densities. For instance, Gaussian smoothing defines

$$p_\sigma := p_0 * \mathcal{N}(0, \sigma^2 I_d), \quad \sigma \geq 0,$$

which can be viewed as a bridge from the data distribution ($\sigma = 0$) toward a Gaussian reference as σ increases. Along this noising family, a natural “DSM time score” is given as

$$s_\sigma^{\text{DSM-time}}(x) := \partial_\sigma \log p_\sigma(x).$$

If we choose the DRE- ∞ bridge to coincide with the DSM noising family up to a smooth reparameterization, i.e. $p_t \equiv p_{\sigma(t)}$ for a monotone C^1 map $t \mapsto \sigma(t)$, then the two time scores represent the same infinitesimal object and differ only by the deterministic change of variables:

$$s_t^{\text{time}}(x) = \partial_t \log p_{\sigma(t)}(x) = \sigma'(t) \partial_\sigma \log p_\sigma(x) \big|_{\sigma=\sigma(t)} = \sigma'(t) s_{\sigma(t)}^{\text{DSM-time}}(x).$$

Consequently, under a shared bridge (and possibly a deterministic reparameterization of time), the “time score” in DRE- ∞ can be interpreted as the same time-derivative-of-log-density object that arises from DSM. In particular, for any σ_0, σ_1 with $0 \leq \sigma_0 < \sigma_1$,

$$\log \frac{p_{\sigma_0}(x)}{p_{\sigma_1}(x)} = \int_{\sigma_1}^{\sigma_0} s_\sigma^{\text{time}}(x) d\sigma = \int_{t_1}^{t_0} s_t^{\text{time}}(x) dt, \quad \text{where } \sigma(t_j) = \sigma_j.$$

D. Implementation Details: Time-ScoreMatchingRiesz

This appendix makes the design of *Time-ScoreMatchingRiesz* (*Time-SMR*) explicit. Time-SMR is a multi-task score-learning module that combines (i) *Data-ScoreMatchingRiesz* (data-score learning via denoising score matching) and (ii) *Time-ScoreMatchingRiesz* (time-score learning via infinitesimal classification / time-score matching; Choi et al., 2022). We specify (a) the parameter-sharing pattern, (b) the joint objective and loss weights, and (c) the training schedule and inference procedure.

D.1. Shared vs. separate networks

Let $x \in \mathcal{X} \subset \mathbb{R}^d$ denote the regressor. Time-SMR involves two score fields:

$$s^{\text{data}}(x, \sigma) \in \mathbb{R}^d, \quad s^{\text{time}}(x, t) \in \mathbb{R},$$

where σ is a DSM noise level and t is the bridge index used in time-score matching.

A reproducible parameterization is a *shared backbone with two heads*:

$$h_\phi(x, u) \in \mathbb{R}^H, \quad s_\theta^{\text{data}}(x, \sigma) = g_{\theta_d}^{\text{data}}(h_\phi(x, \text{emb}(\sigma))), \quad s_\theta^{\text{time}}(x, t) = g_{\theta_t}^{\text{time}}(h_\phi(x, \text{emb}(t))),$$

where ϕ are shared parameters, (θ_d, θ_t) are head-specific parameters, and $\text{emb}(\cdot)$ is a scalar embedding (e.g., Fourier/sinusoidal features). This sharing choice is motivated by the joint-training perspective of Choi et al. (2022), where a single model can represent multiple infinitesimal score components along a continuum.

Alternative (fully separate networks). For maximum modularity, one can use disjoint networks $s_{\theta_d}^{\text{data}}$ and $s_{\theta_t}^{\text{time}}$. All steps below remain identical, except that ϕ is absent. In our experience, sharing a backbone is often more stable when one task (typically data-score learning) is easier and regularizes the representation used by the other task.

D.2. Losses and joint objective

Data-score loss (Data-ScoreMatchingRiesz). We use denoising score matching with Gaussian noise: $\tilde{x} = x + \sigma\varepsilon$, $\varepsilon \sim \mathcal{N}(0, I_d)$, $\sigma \sim \pi$. The per-minibatch loss is

$$\mathcal{L}_{\text{data}}(\theta) = \mathbb{E}_{x \sim p_0} \mathbb{E}_{\sigma \sim \pi} \mathbb{E}_{\varepsilon} \left[\lambda_{\text{data}}(\sigma) \|s_{\theta}^{\text{data}}(x + \sigma\varepsilon, \sigma) + \varepsilon/\sigma\|_2^2 \right]. \quad (3)$$

(For AME estimation with $x = (d, z)$, one may inject noise only in the d -coordinate to directly learn $\partial_d \log p_0(d, z)$; see Appendix L for the corresponding discussion.)

Time-score loss (Time-ScoreMatchingRiesz; Choi et al., 2022). Let q and p be endpoint densities and $\{p_t\}_{t \in [0,1]}$ a differentiable bridge. Let $x_0 \sim q$, $x_1 \sim p$, and $x_t \leftarrow \text{INTERPOLATE}(x_0, x_1, t)$ with $t \sim \text{Unif}(0, 1)$. We minimize the integration-by-parts objective (the continuum limit underlying infinitesimal classification), implemented by Monte Carlo as

$$\hat{\mathcal{L}}_{\text{time}}(\theta) = \underbrace{\lambda(t) \partial_t s_{\theta}^{\text{time}}(x_t, t) + \lambda'(t) s_{\theta}^{\text{time}}(x_t, t) + \frac{1}{2} \lambda(t) s_{\theta}^{\text{time}}(x_t, t)^2}_{\text{integrand term}} + \underbrace{\lambda(0) s_{\theta}^{\text{time}}(x_0, 0) - \lambda(1) s_{\theta}^{\text{time}}(x_1, 1)}_{\text{endpoint terms}}. \quad (4)$$

Here $\partial_t s_{\theta}^{\text{time}}(x_t, t)$ is a *partial* derivative in t holding x_t fixed, computed by automatic differentiation.

Joint objective (loss weighting). Time-SMR minimizes a weighted sum:

$$\mathcal{L}_{\text{joint}}(\theta) = \omega_{\text{data}} \mathcal{L}_{\text{data}}(\theta) + \omega_{\text{time}} \mathcal{L}_{\text{time}}(\theta), \quad (5)$$

with fixed nonnegative weights $(\omega_{\text{data}}, \omega_{\text{time}})$. A simple default is $\omega_{\text{data}} = \omega_{\text{time}} = 1$. To make results reproducible without manual tuning, one can also *normalize* by the initial minibatch scales: set $\omega_{\text{data}} = 1$ and $\omega_{\text{time}} = \mathbb{E}[\hat{\mathcal{L}}_{\text{data}}] / \mathbb{E}[\hat{\mathcal{L}}_{\text{time}}]$ measured on a few initial minibatches, and then keep it fixed.

D.3. Training schedule

Time-SMR can be trained either (i) fully simultaneously or (ii) with a short warm-up. We recommend the following schedule, which is simple and tends to stabilize training:

- **Warm-up (optional).** Train only the data-score module for N_{warm} iterations (i.e., set $\omega_{\text{time}} = 0$).
- **Joint phase.** After warm-up, optimize Eq. (5) with fixed $(\omega_{\text{data}}, \omega_{\text{time}})$. Optionally ramp ω_{time} linearly from 0 to its final value over N_{ramp} iterations.

D.4. Pseudocode

Stabilizers (recommended). When ratios are constructed by exponentiating integrated scores, we recommend: (i) *log-ratio clipping* before exponentiation and (ii) *mean-one calibration* on the training fold (i.e., subtract $\log \mathbb{E}_{\text{train}}[\exp(\log r(X))]$ to enforce $\mathbb{E}_{\text{train}}[\hat{r}(X)] = 1$). These steps reduce extreme weights and improve downstream stability.

E. Application to ATE Estimation

E.1. Riesz Representer in ATE Estimation

Let $X = (D, Z)$ with $D \in \{1, -1\}$ and covariates Z . Let $(Y(1), Y(-1))$ be potential outcomes and $Y = \mathbb{1}[D = 1]Y(1) + \mathbb{1}[D = -1]Y(-1)$. The ATE is

$$\theta_0^{\text{ATE}} := \mathbb{E}[Y(1) - Y(-1)] = \mathbb{E}[\gamma_0(1, Z) - \gamma_0(-1, Z)],$$

Algorithm 1 Time-ScoreMatchingRiesz: one training iteration

Require: Baseline samples $\mathcal{D}_0 \sim p_0$; endpoint samples $\mathcal{D}_q \sim q$, $\mathcal{D}_p \sim p$; noise distribution π ; interpolation routine INTERPOLATE; weights $\omega_{\text{data}}, \omega_{\text{time}}$; weight functions $\lambda_{\text{data}}(\cdot)$ and $\lambda(\cdot)$; iteration index k ; warm-up steps N_{warm} .

- 1: Sample minibatch $\{x\}$ from $\mathcal{D}_0$; sample $\sigma \sim \pi$ and $\varepsilon \sim \mathcal{N}(0, I_d)$; set $\tilde{x} \leftarrow x + \sigma\varepsilon$.
- 2: $\hat{\mathcal{L}}_{\text{data}} \leftarrow \frac{1}{B} \sum \lambda_{\text{data}}(\sigma) \|s_{\theta}^{\text{data}}(\tilde{x}, \sigma) + \varepsilon/\sigma\|_2^2$.
- 3: Sample $t \sim \text{Unif}(0, 1)$; sample $x_0 \sim \mathcal{D}_q$, $x_1 \sim \mathcal{D}_p$; set $x_t \leftarrow \text{INTERPOLATE}(x_0, x_1, t)$.
- 4: Compute $\hat{\mathcal{L}}_{\text{time}}$ via Eq. (4) (using autodiff for $\partial_t s_{\theta}^{\text{time}}(x_t, t)$).
- 5: **if** $k \leq N_{\text{warm}}$ **then**
- 6: $\omega_{\text{time}}^{(k)} \leftarrow 0$
- 7: **else**
- 8: $\omega_{\text{time}}^{(k)} \leftarrow \omega_{\text{time}}$ (or a linear ramp)
- 9: **end if**
- 10: $\hat{\mathcal{L}}_{\text{joint}} \leftarrow \omega_{\text{data}} \hat{\mathcal{L}}_{\text{data}} + \omega_{\text{time}}^{(k)} \hat{\mathcal{L}}_{\text{time}}$.
- 11: Update parameters by one gradient step on $\hat{\mathcal{L}}_{\text{joint}}$.

Algorithm 2 Time-SMR inference: constructing representers from learned scores

Require: Trained $(s_{\theta}^{\text{data}}, s_{\theta}^{\text{time}})$; target type (AME or ratio-based); quadrature rule with M grid points; clipping level c (optional).

- 1: **if** target is AME (data-score-based) **then**
- 2: Choose a small evaluation noise σ_{eval} (e.g. the smallest trained noise).
- 3: Set $\hat{\alpha}_{\text{AME}}(d, z) \leftarrow - (s_{\theta}^{\text{data}}((d, z), \sigma_{\text{eval}}))_d$.
- 4: **else**
- 5: For each x , compute $\widehat{\log r}(x) \leftarrow - \int_0^1 s_{\theta}^{\text{time}}(x, t) dt$ by numerical integration on M grid points.
- 6: Optionally clip: $\widehat{\log r}(x) \leftarrow \text{clip}(\widehat{\log r}(x), [-c, c])$.
- 7: Set $\hat{r}(x) \leftarrow \exp(\widehat{\log r}(x))$ and form $\hat{\alpha}$ from ratios (difference-of-ratios if needed).
- 8: **end if**

where $\gamma_0(d, z) = \mathbb{E}[Y \mid D = d, Z = z]$ under standard identification assumptions (e.g. unconfoundedness and overlap). Let us define

$$m^{\text{ATE}}(W, \gamma) := \gamma(1, Z) - \gamma(-1, Z).$$

Then, the Riesz representer is given as

$$\alpha_0^{\text{ATE}}(D, Z) := \frac{\mathbb{1}[D = 1]}{e_0(Z)} - \frac{\mathbb{1}[D = -1]}{1 - e_0(Z)},$$

where $e_0(z) := P(D = 1 \mid Z = z)$ is the propensity score.

E.2. Time-ScoreMatchingRiesz in ATE Estimation

Let $\pi := P(D = 1)$ and denote the conditional covariate densities $p_1(z) := p_0(z \mid D = 1)$ and $p_{-1}(z) := p_0(z \mid D = -1)$, and the marginal $p_0(z) = \pi p_1(z) + (1 - \pi) p_{-1}(z)$. Then, we have $e_0(z) = P(D = 1 \mid Z = z) = \pi \frac{p_1(z)}{p_0(z)}$. Hence, the ATE Riesz representer is written as $\alpha_0^{\text{ATE}}(D, Z) = \mathbb{1}[D = 1] \frac{p_0(Z)}{\pi p_1(Z)} - \mathbb{1}[D = -1] \frac{p_0(Z)}{(1 - \pi) p_{-1}(Z)}$. Therefore, estimating α_0^{ATE} reduces to estimating the density ratios p_0/p_1 and p_0/p_{-1} .

Data-score-based alternative. A Data-ScoreMatchingRiesz approach could also target α_0^{ATE} by separately estimating the covariate scores for p_0 , p_1 , and p_{-1} and then reconstructing the corresponding log densities. Pursuing this route requires specifying the estimation procedure, the treatment of normalizing constants in the reconstruction, and an error analysis. We therefore do not develop this alternative here.

Bridge construction. Construct a bridge family $\{p_t\}_{t \in [-1, 1]}$ on the Z space as follows. Let $Z_0 \sim p_0$, $Z_1 \sim p_1$, and $Z_{-1} \sim p_{-1}$ be mutually independent. Let $\beta^{(1)}, \beta^{(2)} : [0, 1] \rightarrow [0, 1]$ be C^2 functions satisfying

$$\beta^{(1)}(0) = 1, \beta^{(2)}(0) = 0, \quad \beta^{(1)}(1) = 0, \beta^{(2)}(1) = 1,$$

with (for instance) $\beta^{(1)}$ nonincreasing and $\beta^{(2)}$ nondecreasing. Define

$$Z_t := \begin{cases} \beta^{(1)}(t)Z_0 + \beta^{(2)}(t)Z_1, & t \in [0, 1], \\ \beta^{(1)}(-t)Z_0 + \beta^{(2)}(-t)Z_{-1}, & t \in [-1, 0]. \end{cases}$$

Let p_t denote the induced law of Z_t .

Time score estimation. Let $s_{\theta,t}^{\text{time}}(z)$ approximate $s_t^{\text{time}}(z) = \partial_t \log p_t(z)$ on $t \in (-1, 1)$. We fit s_{θ}^{time} by minimizing the integration-by-parts objective (the natural extension of Section 3):

$$\begin{aligned} \mathcal{R}^{\text{ATE}}(s_{\theta}^{\text{time}}) &:= \mathbb{E}_{Z_{-1} \sim p_{-1}} [\lambda(-1)s_{\theta,-1}^{\text{time}}(Z_{-1})] \\ &\quad - \mathbb{E}_{Z_1 \sim p_1} [\lambda(1)s_{\theta,+1}^{\text{time}}(Z_1)] + \\ &\quad \int_{-1}^1 \mathbb{E}_{Z_t \sim p_t} \left[\partial_t (\lambda(t)s_{\theta,t}^{\text{time}}(Z_t)) + \frac{1}{2} \lambda(t)s_{\theta,t}^{\text{time}}(Z_t)^2 \right] dt. \end{aligned}$$

As discussed earlier, $\partial_t s_{\theta,t}^{\text{time}}(z)$ is a partial derivative in t at fixed z .

Constructing $\hat{\alpha}^{\text{ATE}}$. After training, we obtain $\widehat{\log \frac{p_0}{p_1}}(z) := \int_1^0 s_{\theta,t}^{\text{time}}(z) dt$ and $\widehat{\log \frac{p_0}{p_{-1}}}(z) := \int_{-1}^0 s_{\theta,t}^{\text{time}}(z) dt$. Then, we estimate the Riesz representer as

$$\begin{aligned} \hat{\alpha}^{\text{ATE}}(D, Z) &:= \\ \mathbb{1}[D = 1] \frac{\exp\left(\widehat{\log \frac{p_0}{p_1}}(Z)\right)}{\pi} - \mathbb{1}[D = -1] \frac{\exp\left(\widehat{\log \frac{p_0}{p_{-1}}}(Z)\right)}{1 - \pi}. \end{aligned}$$

Remark (Alternative (logit) parameterization). *One can also estimate $\log(p_1(z)/p_{-1}(z))$ via $\log \frac{p_1(z)}{p_{-1}(z)} \approx \int_{-1}^1 s_{\theta,t}^{\text{time}}(z) dt$ which yields a logistic representation of the propensity score:*

$$e_0(z) \approx \frac{1}{1 + \exp\left(-\left(\log \frac{\pi}{1-\pi} + \int_{-1}^1 s_{\theta,t}^{\text{time}}(z) dt\right)\right)}.$$

F. Policy Path Estimation Algorithm

Algorithm 3 summarizes the estimator in Section 7.

Algorithm 3 Cross-fitted policy path estimation under shift policies

1. Input: data $\{(X_i, Y_i)\}_{i=1}^n$, grid $\{\delta_j\}_{j=1}^J$, K folds.
2. For each fold k :
 - (a) Fit $\hat{\gamma}^{(-k)}$ on training folds; compute residuals $Y_i - \hat{\gamma}^{(-k)}(X_i)$ for i in fold k .
 - (b) Fit a score model on training folds to obtain $\hat{\alpha}^{\text{AME},(-k)}$.
 - (c) For each δ_j :
 - i. Compute $\widehat{\log r_{\pm\delta_j}^{(-k)}}(X_i)$ for test points by integrating $\hat{\alpha}^{\text{AME},(-k)}$ along the D -path.
 - ii. Calibrate $\widehat{r_{\pm\delta_j}^{(-k)}}$ to have mean one on the training fold; set $\hat{\alpha}_{\delta_j}^{(-k)} = \widehat{r_{+\delta_j}^{(-k)}} - \widehat{r_{-\delta_j}^{(-k)}}$.
 - iii. Form orthogonal scores

$$\psi_{i,\delta_j} = \hat{\gamma}^{(-k)}(D_i + \delta_j, Z_i) - \hat{\gamma}^{(-k)}(D_i - \delta_j, Z_i) + \hat{\alpha}_{\delta_j}^{(-k)}(X_i)(Y_i - \hat{\gamma}^{(-k)}(X_i)).$$

3. Output $\hat{\theta}(\delta_j) = n^{-1} \sum_{i=1}^n \psi_{i,\delta_j}$ and pointwise standard errors from $\{\psi_{i,\delta_j}\}_{i=1}^n$.
-

G. Pushforward Policies: Definitions and Data-Score-Based Construction of APEs and Policy Paths

This appendix formalizes pushforward policies and explains why, when the pushforward map is known, both (i) APE representers and (ii) the full policy path can be constructed from the observational *data score*. The key point is that the counterfactual density p_δ is then determined by p_0 and a known map.

G.1. Pushforward of a distribution

Let $(\mathcal{X}, \mathcal{B})$ be a measurable space, P a probability measure on it, and $T : \mathcal{X} \rightarrow \mathcal{X}$ a measurable map. The *pushforward* (a.k.a. image measure) of P by T is the measure $T_\#P$ defined by

$$(T_\#P)(A) := P(T^{-1}(A)), \quad A \in \mathcal{B}.$$

Equivalently, if $X \sim P$, then $T(X) \sim T_\#P$.

Density under a diffeomorphism. When $\mathcal{X} \subseteq \mathbb{R}^d$, P admits a density p w.r.t. Lebesgue measure, and T is a C^1 diffeomorphism, the pushforward $T_\#P$ admits a density

$$p_T(x) = p(T^{-1}(x)) \left| \det \nabla T^{-1}(x) \right|. \quad (6)$$

Equation Eq. (6) is the standard change-of-variables formula.

G.2. Known pushforward policy families

A counterfactual policy family $\{P_\delta\}_{\delta \in \mathbb{R}}$ is a *known pushforward* of P_0 if there exists a *known* family of measurable maps $\{T_\delta : \mathcal{X} \rightarrow \mathcal{X}\}_{\delta \in \mathbb{R}}$ such that

$$P_\delta = (T_\delta)_\#P_0, \quad \text{i.e.} \quad X_\delta = T_\delta(X) \text{ for } X \sim P_0.$$

Assume P_0 has a density p_0 and that each T_δ is a C^1 diffeomorphism. Then P_δ has density

$$p_\delta(x) = p_0(T_\delta^{-1}(x)) \left| \det \nabla T_\delta^{-1}(x) \right|. \quad (7)$$

Define the density ratio

$$r_\delta(x) := \frac{p_\delta(x)}{p_0(x)}.$$

By Eq. (7),

$$\log r_\delta(x) = \underbrace{\log p_0(T_\delta^{-1}(x)) - \log p_0(x)}_{\text{data-score term}} + \underbrace{\log \left| \det \nabla T_\delta^{-1}(x) \right|}_{\text{Jacobian term}}. \quad (8)$$

G.3. From the data score to density ratios under known pushforwards

Let the *data score* be $s_0^{\text{data}}(x) := \nabla_x \log p_0(x)$. The first term in Eq. (8) can be written as a line integral of the data score along the known inverse path $u \mapsto T_u^{-1}(x)$:

$$\log p_0(T_\delta^{-1}(x)) - \log p_0(x) = \int_0^\delta \left\langle s_0^{\text{data}}(T_u^{-1}(x)), \partial_u T_u^{-1}(x) \right\rangle du. \quad (9)$$

Combining Eq. (8) and Eq. (9) yields an explicit data-score-based ratio formula:

$$\log r_\delta(x) = \int_0^\delta \left\langle s_0^{\text{data}}(T_u^{-1}(x)), \partial_u T_u^{-1}(x) \right\rangle du + \log \left| \det \nabla T_\delta^{-1}(x) \right|. \quad (10)$$

Thus, once we estimate the data score s_0^{data} on P_0 , the entire family $\{r_\delta\}_\delta$ is available by (i) evaluating the learned score along the known path and (ii) adding the known Jacobian term.

Translation (shift) policy. Let $X = (D, Z)$ and $T_\delta(d, z) = (d + \delta, z)$. Then $T_\delta^{-1}(d, z) = (d - \delta, z)$ and $|\det \nabla T_\delta^{-1}| \equiv 1$, so

$$\log r_\delta(d, z) = \log p_0(d - \delta, z) - \log p_0(d, z) = - \int_0^\delta \partial_d \log p_0(d - u, z) du. \quad (11)$$

Recalling the AME representer $\alpha_0^{\text{AME}}(d, z) = -\partial_d \log p_0(d, z)$, we obtain

$$\log r_\delta(d, z) = \int_0^\delta \alpha_0^{\text{AME}}(d - u, z) du, \quad r_\delta(d, z) = \exp \left(\int_0^\delta \alpha_0^{\text{AME}}(d - u, z) du \right). \quad (12)$$

This is the basic “score-to-ratio” identity used by Data-ScoreMatchingRiesz in the known-pushforward regime.

G.4. APE estimation under known pushforwards from the data score

Fix two counterfactual indices δ_1, δ_2 and define the APE

$$\theta_0(\delta_1, \delta_2) := \mathbb{E}_{X \sim p_{\delta_1}} [\gamma_0(X)] - \mathbb{E}_{X \sim p_{\delta_2}} [\gamma_0(X)].$$

When $p_{\delta_j} \ll p_0$, the APE has the density-ratio form

$$\theta_0(\delta_1, \delta_2) = \mathbb{E} \left[\alpha_{0, \delta_1, \delta_2}(X) \gamma_0(X) \right], \quad \alpha_{0, \delta_1, \delta_2}(x) := r_{\delta_1}(x) - r_{\delta_2}(x).$$

Under a known pushforward, each ratio r_{δ_j} can be constructed from the data score via Eq. (10) (or Eq. (12) for translations), so the APE representer $\alpha_{0, \delta_1, \delta_2}$ is identified from the observational score and known policy maps.

Orthogonal-score estimation (generic form). Under a known pushforward $X_\delta = T_\delta(X)$, the APE admits the moment representation

$$\theta_0(\delta_1, \delta_2) = \mathbb{E}[\gamma_0(T_{\delta_1}(X)) - \gamma_0(T_{\delta_2}(X))] \equiv \mathbb{E}[m_{\delta_1, \delta_2}(W, \gamma)], \quad m_{\delta_1, \delta_2}(W, \gamma) := \gamma(T_{\delta_1}(X)) - \gamma(T_{\delta_2}(X)).$$

Let $\alpha_{0, \delta_1, \delta_2}(x) = r_{\delta_1}(x) - r_{\delta_2}(x)$ denote the corresponding Riesz representer, so that $\mathbb{E}[m_{\delta_1, \delta_2}(W, \gamma)] = \mathbb{E}[\alpha_{0, \delta_1, \delta_2}(X) \gamma(X)]$ for all γ . An orthogonal-score estimator is then

$$\hat{\theta}(\delta_1, \delta_2) = \frac{1}{n} \sum_{i=1}^n \left[m_{\delta_1, \delta_2}(W_i, \hat{\gamma}) + \hat{\alpha}_{\delta_1, \delta_2}(X_i) (Y_i - \hat{\gamma}(X_i)) \right],$$

with $\hat{\alpha}_{\delta_1, \delta_2}$ constructed from the learned score via $\hat{r}_\delta$ and $\hat{\alpha}_{\delta_1, \delta_2} = \hat{r}_{\delta_1} - \hat{r}_{\delta_2}$. Cross-fitted mean-one calibration and log clipping are recommended for stability (Appendix M.2).

G.5. Policy-path estimation under known pushforwards

The policy path in the main text is the symmetric APE curve

$$\theta_0(\delta) \equiv \theta_0(+\delta, -\delta), \quad \alpha_{0, \delta}(x) = r_{+\delta}(x) - r_{-\delta}(x).$$

Under known pushforwards, we can estimate the *entire curve* $\delta \mapsto \theta_0(\delta)$ by reusing a *single* data-score estimate of p_0 : for each δ on a grid, compute $\hat{r}_{\pm\delta}$ from the learned score (plus known Jacobian), form $\hat{\alpha}_{0, \delta} = \hat{r}_{+\delta} - \hat{r}_{-\delta}$, and plug into the orthogonal score.

Shift policy and the AME limit (revisited). For translation policies, Eq. (12) implies that the full path representer $\alpha_{0, \delta}$ is constructed from the AME representer by one-dimensional score integration. Moreover, the path itself satisfies

$$\theta_0(\delta) = \mathbb{E}[\gamma_0(D + \delta, Z) - \gamma_0(D - \delta, Z)],$$

hence $\theta'_0(0) = 2\mathbb{E}[\partial_d \gamma_0(D, Z)] = 2\theta_0^{\text{AME}}$ under mild smoothness (Section 7). This identity generally fails without a known pushforward structure.

Implementation. A cross-fitted algorithm that estimates $\delta \mapsto \theta_0(\delta)$ on a grid under shift policies is given in Appendix F. The only pushforward-specific step is the construction of $\hat{r}_{\pm\delta}$ from the learned data score, using Eq. (12) (or Eq. (10) for general diffeomorphisms).

H. Proof of Theorem 6.1

Proof. Expand the definition of $\text{BD}_g^\dagger$ and drop the term

$$\int_0^1 \int \lambda(t) g(\partial_t \log p_t(x)) p_t(x) dx dt,$$

which does not depend on θ . The remaining θ -dependent part is

$$\begin{aligned} \text{BD}_g^\dagger(\partial_t \log p_t \| s_\theta^{\text{time}}) &= \int_0^1 \mathbb{E}_{X_t \sim p_t} \left[\lambda(t) \left(g'(s_\theta^{\text{time}}(X_t, t)) s_\theta^{\text{time}}(X_t, t) - g(s_\theta^{\text{time}}(X_t, t)) \right) \right] dt \\ &\quad - \int_0^1 \int \lambda(t) g'(s_{\theta,t}^{\text{time}}(x)) \partial_t \log p_t(x) p_t(x) dx dt. \end{aligned}$$

Using $\partial_t \log p_t(x) p_t(x) = \partial_t p_t(x)$, the second term becomes

$$- \int_0^1 \int \lambda(t) g'(s_{\theta,t}^{\text{time}}(x)) \partial_t p_t(x) dx dt.$$

Apply integration by parts in t (for each fixed x):

$$\begin{aligned} - \int_0^1 \lambda(t) g'(s_{\theta,t}^{\text{time}}(x)) \partial_t p_t(x) dt &= - \left[\lambda(t) g'(s_{\theta,t}^{\text{time}}(x)) p_t(x) \right]_{t=0}^{t=1} \\ &\quad + \int_0^1 \partial_t \left(\lambda(t) g'(s_{\theta,t}^{\text{time}}(x)) \right) p_t(x) dt \\ &= \lambda(0) g'(s_\theta^{\text{time}}(x, 0)) p_0(x) - \lambda(1) g'(s_\theta^{\text{time}}(x, 1)) p_1(x) \\ &\quad + \int_0^1 \left(\lambda(t) \partial_t g'(s_{\theta,t}^{\text{time}}(x)) + \lambda'(t) g'(s_{\theta,t}^{\text{time}}(x)) \right) p_t(x) dt. \end{aligned}$$

Integrating over x and combining terms yields exactly $\text{BD}_g(s_\theta^{\text{time}})$ as stated in Theorem 6.1, up to constants independent of θ . $\square$

I. Additional Distribution-Estimation Metrics for Diffusion Models

This appendix section records classical distribution-estimation guarantees for diffusion models (in metrics other than the L_2 score norms used in the main text), as established by [Oko et al. \(2023\)](#).

Total variation distance. For probability measures μ, ν on $\mathcal{X}$ with densities p, q , define

$$\text{TV}(\mu, \nu) := \sup_{A \subseteq \mathcal{X}} |\mu(A) - \nu(A)| = \frac{1}{2} \int_{\mathcal{X}} |p(x) - q(x)| dx.$$

Wasserstein-1 distance. Define

$$W_1(\mu, \nu) := \inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{X}} \|x - y\|_2 \pi(dx, dy),$$

where $\Pi(\mu, \nu)$ is the set of couplings of (μ, ν) .

Theorem I.1 (Total variation rate; [Oko et al. \(2023\)](#)). *Under the assumptions of [Oko et al. \(2023\)](#), the diffusion-model density estimator $\hat{\mu}$ constructed by score matching satisfies*

$$\mathbb{E}[\text{TV}(\hat{\mu}, p_0)] \lesssim n^{-s/(2s+d)} \cdot \text{polylog}(n).$$

Moreover, the minimax lower bound over the Besov class matches $n^{-s/(2s+d)}$ up to logarithmic factors.

Theorem I.2 (Wasserstein-1 rate; [Oko et al. \(2023\)](#)). *Under the assumptions of [Oko et al. \(2023\)](#), for any fixed $\delta > 0$ the diffusion-model estimator can be constructed so that*

$$\mathbb{E}[W_1(\hat{\mu}, p_0)] \lesssim n^{-(s+1-\delta)/(2s+d)}.$$

A minimax lower bound over Besov classes is $n^{-(s+1)/(2s+d)}$ (hence the rate above is nearly minimax optimal).

J. Proof of Theorem 8.2

We now derive a generic nonparametric convergence rate for the *time-score matching* estimator used in ScoreMatchingRiesz (DRE- ∞).

J.1. Setup: bridge distributions and time scores

Let q and p be two densities on $\mathcal{X} \subseteq \mathbb{R}^d$ with $q \ll p$, and define the density ratio $r(x) := q(x)/p(x)$. Following the DRE- ∞ construction, introduce a continuum of bridge laws $\{p_t\}_{t \in [0,1]}$ connecting q and p with $p_0 = q$ and $p_1 = p$, and define the *time score*

$$s_{0,t}(x) := \partial_t \log p_t(x).$$

The key identity is the continuum telescoping representation

$$\log r(x) = \log p_0(x) - \log p_1(x) = \int_1^0 s_{0,t}(x) dt = - \int_0^1 s_{0,t}(x) dt.$$

We consider the weighted L_2 -norm on functions $f: \mathcal{X} \times [0, 1] \rightarrow \mathbb{R}$:

$$\|f\|_{\mu_\lambda}^2 := \int_0^1 \mathbb{E}_{X_t \sim p_t} [\lambda(t) f(X_t, t)^2] dt, \quad \lambda(t) > 0.$$

J.2. Population objective and quadratic identity

Let $s: \mathcal{X} \times [0, 1] \rightarrow \mathbb{R}$ be a differentiable candidate time-score function. Consider the integration-by-parts population objective

$$\begin{aligned} \mathcal{R}(s) &:= \mathbb{E}_{X_0 \sim q} [\lambda(0) s(X_0, 0)] - \mathbb{E}_{X_1 \sim p} [\lambda(1) s(X_1, 1)] \\ &\quad + \int_0^1 \mathbb{E}_{X_t \sim p_t} \left[\partial_t (\lambda(t) s(X_t, t)) + \frac{1}{2} \lambda(t) s(X_t, t)^2 \right] dt. \end{aligned} \quad (13)$$

Lemma J.1 (Quadratic identity). *Assume $t \mapsto p_t(x)$ is differentiable for a.e. x , s is differentiable in t , and boundary terms are well-defined. Then, up to an additive constant independent of s ,*

$$\mathcal{R}(s) = \mathcal{R}(s_0) + \frac{1}{2} \|s - s_0\|_{\mu_\lambda}^2.$$

In particular, s_0 is the unique population minimizer of $\mathcal{R}(\cdot)$ over any function class that contains s_0 .

Proof. Expand the weighted squared loss:

$$\frac{1}{2} \|s - s_0\|_{\mu_\lambda}^2 = \frac{1}{2} \int_0^1 \mathbb{E}_{X_t \sim p_t} [\lambda(t) s^2] dt - \int_0^1 \mathbb{E}_{X_t \sim p_t} [\lambda(t) s s_0] dt + \frac{1}{2} \int_0^1 \mathbb{E}_{X_t \sim p_t} [\lambda(t) s_0^2] dt.$$

The last term does not depend on s and can be absorbed into a constant.

For the cross term, use $s_{0,t}(x) = \partial_t \log p_t(x)$ so that $s_{0,t}(x) p_t(x) = \partial_t p_t(x)$, hence

$$\mathbb{E}_{X_t \sim p_t} [\lambda(t) s(X_t, t) s_{0,t}(X_t)] = \int \lambda(t) s(x, t) \partial_t p_t(x) dx.$$

Integrate by parts in t (for each fixed x) to obtain

$$\int_0^1 \lambda(t) s(x, t) \partial_t p_t(x) dt = \left[\lambda(t) s(x, t) p_t(x) \right]_{t=0}^{t=1} - \int_0^1 \partial_t (\lambda(t) s(x, t)) p_t(x) dt.$$

Integrating over x yields

$$\int_0^1 \mathbb{E}_{X_t \sim p_t} [\lambda(t) s s_0] dt = \mathbb{E}_{X_1 \sim p} [\lambda(1) s(X_1, 1)] - \mathbb{E}_{X_0 \sim q} [\lambda(0) s(X_0, 0)]$$

$$- \int_0^1 \mathbb{E}_{X_t \sim p_t} \left[\partial_t (\lambda(t) s(X_t, t)) \right] dt.$$

Substituting this expression into the expansion of $\frac{1}{2} \|s - s_0\|_{\mu_\lambda}^2$ shows that $\mathcal{R}(s)$ in Eq. (13) equals $\mathcal{R}(s_0) + \frac{1}{2} \|s - s_0\|_{\mu_\lambda}^2$, up to constants independent of s . $\square$

J.3. Proof of Theorem 8.2

Proof. By definition of $\widehat{s}$ and the basic inequality,

$$\widehat{\mathcal{R}}_n(\widehat{s}) \leq \widehat{\mathcal{R}}_n(s) \quad \forall s \in \mathcal{F}_n.$$

Add and subtract the population objective:

$$\mathcal{R}(\widehat{s}) \leq \widehat{\mathcal{R}}_n(\widehat{s}) + \Delta_n \leq \widehat{\mathcal{R}}_n(s) + \Delta_n \leq \mathcal{R}(s) + 2\Delta_n \quad \forall s \in \mathcal{F}_n,$$

hence

$$\mathcal{R}(\widehat{s}) - \mathcal{R}(s_0) \leq \inf_{s \in \mathcal{F}_n} \{\mathcal{R}(s) - \mathcal{R}(s_0)\} + 2\Delta_n.$$

Using Lemma J.1,

$$\mathcal{R}(\widehat{s}) - \mathcal{R}(s_0) = \frac{1}{2} \|\widehat{s} - s_0\|_{\mu_\lambda}^2, \quad \mathcal{R}(s) - \mathcal{R}(s_0) = \frac{1}{2} \|s - s_0\|_{\mu_\lambda}^2,$$

which gives

$$\|\widehat{s} - s_0\|_{\mu_\lambda}^2 \leq \inf_{s \in \mathcal{F}_n} \|s - s_0\|_{\mu_\lambda}^2 + 4\Delta_n.$$

This proves the oracle inequality.

Under Assumption 8.2, the approximation term is $\lesssim P_n^{-2s/(d+1)}$, and under Assumption 8.3, $\Delta_n \lesssim P_n \text{polylog}(n)/n$. Balancing the two terms yields the stated rate after taking square roots. $\square$

K. Downstream L_2 Bounds: Log Ratios, Ratios, and Ratio-Based Representers

This section records how L_2 error in an estimated score field propagates to L_2 error of integrated log ratios, ratios, and ratio-based representers. It applies to Time-ScoreMatchingRiesz, and it also applies whenever $\widehat{\log r}$ is constructed as an integral of an estimated score along a known path.

Plug-in log ratio estimator and clipping. Define the plug-in log ratio estimator

$$\widehat{\log r}(x) := \int_1^0 \widehat{s}_t(x) dt = - \int_0^1 \widehat{s}_t(x) dt.$$

When exponentiating, we may apply clipping: for a constant $c > 0$, define $\text{clip}(u, [-c, c]) := \max\{-c, \min\{u, c\}\}$ and set

$$\widehat{r}(x) := \exp(\text{clip}(\widehat{\log r}(x), [-c, c])).$$

Corollary K.1 (Log-ratio and ratio bounds in $L_2(p)$ (with clipping)). *Under Assumption 8.1 and $\lambda(t) \equiv 1$, we have*

$$\|\widehat{\log r} - \log r\|_{L_2(p)} \lesssim \|\widehat{s} - s_0\|_{\mu_1},$$

where $\|g\|_{L_2(p)}^2 := \mathbb{E}_{X \sim p}[g(X)^2]$. Assume additionally that $|\log r(x)| \leq B$ almost surely. If we clip $\widehat{\log r}$ to $[-B, B]$ and set $\widehat{r}(x) := \exp(\text{clip}(\widehat{\log r}(x), [-B, B]))$, then

$$\|\widehat{r} - r\|_{L_2(p)} \lesssim e^B \|\widehat{\log r} - \log r\|_{L_2(p)}.$$

Consequently, under the rate in Theorem 8.2,

$$\|\widehat{r} - r\|_{L_2(p)} = O_{\mathbb{P}}\left(n^{-s/(2s+d+1)} \cdot \text{polylog}(n)\right).$$

Corollary K.2 (APE and Policy-Path Representer Bound in $L_2(p_0)$). *Let p_0 be a baseline density on $\mathcal{X}$ and let p_+ and p_- be two counterfactual densities on $\mathcal{X}$ that are absolutely continuous with respect to p_0 . Define the corresponding density ratios*

$$r_+(x) := \frac{p_+(x)}{p_0(x)}, \quad r_-(x) := \frac{p_-(x)}{p_0(x)},$$

and the ratio-based representer

$$\alpha_0(x) := r_+(x) - r_-(x).$$

Let $\hat{r}_+$ and $\hat{r}_-$ be estimators obtained by score integration (possibly using different bridges), and define

$$\hat{\alpha}(x) := \hat{r}_+(x) - \hat{r}_-(x).$$

If each ratio estimator satisfies $\|\hat{r}_\pm - r_\pm\|_{L_2(p_0)} = O_{\mathbb{P}}(\varepsilon_n)$, then

$$\|\hat{\alpha} - \alpha_0\|_{L_2(p_0)} = O_{\mathbb{P}}(\varepsilon_n).$$

In particular, if Theorem 8.2 applies to both ratio estimators (with $p = p_0$ in Corollary K.1), then one may take $\varepsilon_n = n^{-s/(2s+d+1)} \text{polylog}(n)$.

K.1. Proof of Corollary K.1

Proof of Corollary K.1. For any x ,

$$\widehat{\log r}(x) - \log r(x) = - \int_0^1 (\hat{s}_t(x) - s_{0,t}(x)) dt.$$

By Cauchy–Schwarz,

$$(\widehat{\log r}(x) - \log r(x))^2 \leq \int_0^1 (\hat{s}_t(x) - s_{0,t}(x))^2 dt.$$

Take expectation under $X \sim p$ and use Assumption 8.1 with $p \leq c^{-1}p_t$:

$$\begin{aligned} \|\widehat{\log r} - \log r\|_{L_2(p)}^2 &= \mathbb{E}_{X \sim p} \left[(\widehat{\log r}(X) - \log r(X))^2 \right] \\ &\leq \int_0^1 \mathbb{E}_{X \sim p} \left[(\hat{s}_t(X) - s_{0,t}(X))^2 \right] dt \\ &\leq \frac{1}{c} \int_0^1 \mathbb{E}_{X_t \sim p_t} \left[(\hat{s}_t(X_t) - s_{0,t}(X_t))^2 \right] dt = \frac{1}{c} \|\hat{s} - s_0\|_{\mu_1}^2. \end{aligned}$$

This proves the first inequality.

For the ratio, if $|\log r(x)| \leq B$ and $\widehat{\log r}$ is clipped to $[-B, B]$, then both log ratios lie in $[-B, B]$. Since the map $u \mapsto e^u$ is e^B -Lipschitz on this interval, it follows that

$$|\hat{r}(x) - r(x)| \leq e^B |\widehat{\log r}(x) - \log r(x)|.$$

The $L_2(p)$ bound follows by squaring and integrating. □

K.2. Proof of Corollary K.2

Proof of Corollary K.2. Immediate from the triangle inequality:

$$\|\hat{\alpha} - \alpha_0\|_{L_2(p_0)} \leq \|\hat{r}_+ - r_+\|_{L_2(p_0)} + \|\hat{r}_- - r_-\|_{L_2(p_0)}.$$

□

Optional remark (finite many δ 's). If the score model is trained conditionally in a finite set of policy parameters $\delta \in \{\delta_1, \dots, \delta_M\}$, one can apply a union bound to obtain simultaneous rates over all δ_m at the price of an additional $\log M$ factor in the complexity term.

L. From Data-ScoreMatchingRiesz to the AME Representer: Smoothing Bias and DML Conditions

This appendix addresses the concern that Data-ScoreMatchingRiesz learns a *smoothed* score while the AME Riesz representer involves the *unsmoothed* score of p_0 . We write the discussion in terms of Data-ScoreMatchingRiesz and Time-ScoreMatchingRiesz, rather than DSM and DRE- ∞ .

L.1. What Data-ScoreMatchingRiesz estimates

Let p_0 be the density of $X \in \mathbb{R}^d$. For $\sigma > 0$, define the Gaussian-smoothed density $p_\sigma = p_0 * \varphi_\sigma$ and its data score $s_\sigma(x) := \nabla_x \log p_\sigma(x)$. Data-ScoreMatchingRiesz produces $\hat{s}_\sigma(\cdot)$ that targets $s_\sigma(\cdot)$. For AME with $X = (D, Z)$, the Riesz representer is

$$\alpha_0^{\text{AME}}(d, z) = -\partial_d \log p_0(d, z) = -(s_0(d, z))_d.$$

Our implementation uses a small evaluation noise σ_{eval} and sets $\hat{\alpha}_{\text{AME}}(d, z) := -(\hat{s}_{\sigma_{\text{eval}}}(d, z))_d$.

L.2. Error decomposition: estimation error + smoothing bias

A direct decomposition is

$$\|\hat{\alpha}_{\text{AME}} - \alpha_0^{\text{AME}}\|_{L_2(P_0)} \leq \underbrace{\|(\hat{s}_{\sigma_{\text{eval}}} - s_{\sigma_{\text{eval}}})_d\|_{L_2(P_0)}}_{\text{(A) score estimation error}} + \underbrace{\|(s_{\sigma_{\text{eval}}} - s_0)_d\|_{L_2(P_0)}}_{\text{(B) smoothing bias}}. \quad (14)$$

(A) Estimation error controlled by [Oko et al. \(2023\)](#). Existing diffusion/score-matching theory provides L_2 -type guarantees for estimating s_σ . In particular, under Besov-type smoothness, [Oko et al. \(2023\)](#) gives (near-)minimax rates for score estimation of Gaussian-smoothed densities, which we directly use as a bound on term (A).

(B) Smoothing bias requires a separate (and standard) control. Term (B) is a deterministic approximation error induced by Gaussian smoothing. Under mild regularity (smoothness of p_0 and a lower bound away from zero), $s_\sigma(x) \rightarrow s_0(x)$ as $\sigma \downarrow 0$ and the bias decays polynomially in σ .

Lemma L.1 (Smoothing bias of the data score (illustrative statement)). *Assume p_0 is supported on a bounded domain, satisfies $\inf_x p_0(x) \geq c > 0$, and has smoothness $s > 1$ (e.g., $p_0 \in B_{2,2}^s$ with bounded first derivative). Then there exists $\beta > 0$ (typically $\beta \approx s - 1$) and a constant C such that, for all sufficiently small σ ,*

$$\|s_\sigma - s_0\|_{L_2(P_0)} \leq C \sigma^\beta.$$

Consequently, the AME representer bias satisfies $\|(s_\sigma - s_0)_d\|_{L_2(P_0)} \leq C \sigma^\beta$.

Lemma L.1 is a standard consequence of kernel-approximation bounds for p_σ and ∇p_σ , combined with the identity $s_\sigma = \nabla p_\sigma / p_\sigma$ and the lower bound on p_0 .

L.3. Implications for DML rate conditions and choice of σ_{eval}

Combining Eq. (14) with Lemma L.1 yields

$$\|\hat{\alpha}_{\text{AME}} - \alpha_0^{\text{AME}}\|_{L_2(P_0)} \lesssim \underbrace{\|(\hat{s}_{\sigma_{\text{eval}}} - s_{\sigma_{\text{eval}}})_d\|_{L_2(P_0)}}_{\text{controlled by \a href{Oko et al., 2023}}} + \underbrace{\sigma_{\text{eval}}^\beta}_{\text{smoothing bias}}.$$

For debiased ML with an orthogonal score, a sufficient condition is the product rate

$$\|\hat{\alpha} - \alpha_0\|_{L_2(P_0)} \cdot \|\hat{\gamma} - \gamma_0\|_{L_2(P_0)} = o_p(n^{-1/2}).$$

Thus, σ_{eval} should be chosen so that the smoothing bias term is negligible at the DML scale. For example, if $\|\hat{\gamma} - \gamma_0\|_{L_2} = O_p(n^{-b})$ for some $b > 0$, it suffices that $\sigma_{\text{eval}}^\beta = o(n^{-1/2+b})$. This is the usual *undersmoothing* principle: decrease σ_{eval} with n to kill the smoothing bias faster than the DML remainder.

L.4. Why this issue is specific to Data-ScoreMatchingRiesz (and not Time-ScoreMatchingRiesz)

Time-ScoreMatchingRiesz does not introduce Gaussian smoothing of p_0 ; it estimates a *time score* $s_t^{\text{time}}(x) = \partial_t \log p_t(x)$ along an explicit bridge. The representer is then obtained by integrating the learned time score and exponentiating to form ratios. Accordingly, the analogue of Eq. (14) is an *error propagation* bound from time-score estimation error to log-ratio and ratio error. These bounds are controlled by L_2 guarantees for s^{time} together with stabilization (e.g., clipping and mean-one calibration) when exponentiating.

M. Remarks on the Implementation of ScoreMatchingRiesz

M.1. Infinitesimal classification viewpoint

Time-ScoreMatchingRiesz is closely related to *infinitesimal classification* for DRE. Choi et al. (2022) interprets time score matching as the continuum limit of telescoping density-ratio estimation along a bridge of distributions $\{p_t\}_{t \in [0,1]}$, where one learns the instantaneous change rate $\partial_t \log p_t(x)$.

For small Δt , the Bayes-optimal classifier that discriminates samples from p_t and $p_{t+\Delta t}$ admits a first-order expansion around $1/2$, whose leading term is proportional to the time score. Moreover, letting $\Delta t \rightarrow 0$ in the binary cross-entropy objective yields a squared-error objective equivalent to time score matching. Thus, rather than training many separate classifiers, we learn a single time-score field and recover endpoint log-ratios by integrating it along the bridge.

We exploit the same mechanism for Riesz representer estimation. In many causal targets, the representer can be expressed via endpoint density ratios; our bridge construction converts global ratio estimation into local time-score learning, whose aggregation yields the density ratio and hence the representer used in orthogonal-score estimation.

M.2. Ratio stabilization: cross-fitted calibration and clipping

When ratios are constructed by exponentiating estimated log ratios, both multiplicative constants and numerical instability can degrade downstream estimation. We apply two stabilizers, computed *only on the training fold* and then used for test-fold evaluation:

1. **Cross-fitted mean-one calibration.** Given $\widehat{\ell}(x) = \widehat{\log r}(x)$, define

$$\widehat{\ell}^{\text{cal}}(x) := \widehat{\ell}(x) - \log \mathbb{E}_{\text{train}} [\exp\{\widehat{\ell}(X)\}], \quad \widehat{r}^{\text{cal}}(x) := \exp\{\widehat{\ell}^{\text{cal}}(x)\},$$

so that $\mathbb{E}_{\text{train}}[\widehat{r}^{\text{cal}}(X)] = 1$.

2. **Log clipping before exponentiation.** For a moderate constant $c > 0$, clip

$$\widehat{\ell}^{\text{clip}}(x) := \text{clip}\left(\widehat{\ell}^{\text{cal}}(x), [-c, c]\right), \quad \widehat{r}(x) := \exp\{\widehat{\ell}^{\text{clip}}(x)\}.$$

M.3. Bridge design and overlap in practice

Assumption 8.1 is a convenient sufficient condition for analyzing Time-ScoreMatchingRiesz, but it can be demanding when endpoint distributions are far apart. The following choices make overlap along the bridge more plausible and stabilize the implied weights:

1. **Smooth interpolation schedule β .** For interpolation bridges $X_t = \beta^{(1)}(t)X_0 + \beta^{(2)}(t)X_1$, avoid rapid changes near $t = 0$ and $t = 1$ to prevent unstable endpoint behavior. A simple choice is the cubic schedule

$$\beta^{(2)}(t) = 3t^2 - 2t^3, \quad \beta^{(1)}(t) = 1 - \beta^{(2)}(t),$$

which preserves $\beta^{(2)}(0) = 0, \beta^{(2)}(1) = 1$ while flattening endpoint derivatives.

2. **Endpoint downweighting via λ .** Downweight endpoints by choosing $\lambda(t)$ small near $t = 0$ and $t = 1$ (e.g., $\lambda(t) = t(1-t)$), or equivalently sample t away from endpoints and treat endpoint expectations separately.
3. **If overlap is weak.** If exponentiated ratios become heavy-tailed, add mild Gaussian noise along the bridge for interior t (keeping noise close to zero near endpoints) and apply ratio-stage stabilization via log clipping and mean-one calibration (Appendix M.2).

A simple diagnostic. Report (i) the fraction of test-fold points affected by log-ratio clipping and (ii) a high quantile of the implied weights. Large values typically indicate weak overlap under the chosen bridge and/or insufficient stabilization.

M.4. Stochastic policy intervention (non-pushforward APE)

This appendix specifies the stochastic policy intervention used to illustrate a regime where data-score-based APE construction is not available. Write $X = (D, Z)$ with $Z \in \mathbb{R}^2$ and let P_0 denote the observational law. The policy keeps the marginal law of Z fixed at $p_0(z)$ but replaces the conditional law of $D \mid Z = z$ by a new Gaussian distribution:

$$D^{(+)} \mid Z = z \sim \mathcal{N}(m_+(z), s_+(z)^2), \quad D^{(-)} \mid Z = z \sim \mathcal{N}(m_-(z), s_-(z)^2),$$

where $m_{\pm}$ and $s_{\pm}$ are nonlinear functions of z (chosen to induce substantial but smooth policy changes while maintaining overlap). The resulting counterfactual densities satisfy $p_{\pm}(d, z) = p_0(z) p_{\pm}(d \mid z)$ and are generally not representable as $p_0(T^{-1}(d, z))$ for any known deterministic map T . Consequently, $\nabla \log p_0(d, z)$ does not determine $p_{\pm}/p_0$, so diffusion-style score matching on P_0 alone cannot identify $\alpha^{\text{APE}}(x) = p_+(x)/p_0(x) - p_-(x)/p_0(x)$. By contrast, DRE- ∞ applies directly by treating $(q, p) = (p_{\pm}, p_0)$ and using simulated samples from $p_{\pm}$ together with observed samples from p_0 .

M.5. Practical cautions for policy-path estimation

Policy-path estimation involves repeated ratio construction across δ , which can magnify error:

- **Exponentiation and overlap.** Exponentiating integrated scores can amplify small systematic errors, and limited overlap between p_0 and $p_{\pm\delta}$ typically worsens as $|\delta|$ grows.
- **Stabilization is often essential.** Mean-one calibration and conservative log clipping can materially improve stability of $\theta(\delta)$ estimation (Appendix M.2).
- **Bridge and weighting choices matter (time scores).** Downweighting endpoints and avoiding degenerate endpoint behavior are practical safeguards when estimating ratios along a bridge (Appendix M.3).

These issues are especially visible when estimating an entire curve $\delta \mapsto \theta(\delta)$, rather than a single APE at a fixed δ .

N. Bregman–Riesz regression: feasible Bregman risks for representer estimation

This appendix records a Bregman-divergence formulation of *generalized Riesz regression* due to Kato (2026d). The formulation is useful for viewing a range of representer estimators (including classical Riesz regression and score-matching-type objectives) through a single optimization lens.

Riesz representer identity. Recall that the Riesz representer $\alpha_0 \in L_2(P_X)$ associated with the linear functional $\gamma \mapsto \mathbb{E}[m(W, \gamma)]$ satisfies

$$\mathbb{E}[\alpha_0(X) \gamma(X)] = \mathbb{E}[m(W, \gamma)], \quad \forall \gamma \in \Gamma, \quad (15)$$

for a suitable class $\Gamma \subseteq L_2(P_X)$.

From an oracle Bregman risk to a feasible objective. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be convex and differentiable, and define the scalar Bregman divergence

$$D_g(u, v) := g(u) - g(v) - g'(v)(u - v).$$

A natural “oracle” criterion is the Bregman risk

$$\mathcal{B}_g^\dagger(\alpha) := \mathbb{E}[D_g(\alpha_0(X), \alpha(X))], \quad (16)$$

which is minimized at $\alpha = \alpha_0$ (uniquely if g is strictly convex). Although Eq. (16) involves the unknown α_0 , Kato (2026d) shows that it is equivalent—up to an additive constant independent of α —to the *feasible* objective

$$\mathcal{B}_g(\alpha) := \mathbb{E}\left[-g(\alpha(X)) + g'(\alpha(X))\alpha(X) - m(W, g'(\alpha))\right], \quad (17)$$

where $g'(\alpha)$ denotes the composition $x \mapsto g'(\alpha(x))$. Indeed, expanding Eq. (16) and using Eq. (15) with $\gamma = g'(\alpha)$ yields

$$\mathcal{B}_g^\dagger(\alpha) = \mathbb{E}[g(\alpha_0(X))] + \mathcal{B}_g(\alpha),$$

so minimizing Eq. (17) targets α_0 without evaluating α_0 .

Empirical risk minimization for a sample analog of Eq. (17), with regularization and a model class for α , yields *generalized Riesz regression* (Kato, 2026d). The generator g (loss) and the parameterization of α (link/model) jointly determine which representers are favored. This “loss–link” viewpoint is used in Section 6 when interpreting score-based constructions.

Quadratic generator and classical Riesz regression. For $g(u) = \frac{1}{2}u^2$ we have $g'(u) = u$ and Eq. (17) reduces to

$$\mathcal{B}_{\text{SQ}}(\alpha) = \mathbb{E}\left[\frac{1}{2}\alpha(X)^2 - m(W, \alpha)\right], \quad (18)$$

the population objective underlying (regularized) Riesz regression in automatic debiasing (Chernozhukov et al., 2021; 2022b). In derivative-functionals such as AME, where the representer is a (negative) score, Eq. (18) is also the population Hyvärinen score-matching objective; see Section 3 and Appendix B.

Nonquadratic generators and classical weighting objectives. When the target representer is ratio-based (ATE, APE, policy-path points), α_0 is (a signed combination of) density ratios and therefore a balancing weight. With exponential/logistic-type links, KL-type generators recover calibrated weighting objectives that are classical in covariate balancing and density-ratio estimation (e.g., tailored loss minimization and entropy-style balancing) (Zhao, 2019; Sugiyama et al., 2012; Kato, 2026d). Appendix O records the relevant balancing identities and their role in controlling the Neyman remainder.

Relation to Bregman and generalized score matching. Beyond the quadratic case, there is a broader literature on Bregman-divergence formulations of score/ratio estimation. For unnormalized models, Gutmann & Hirayama (2011) embed score matching in a larger Bregman framework; Lyu (2009) generalizes score matching via linear operators, yielding generalized Fisher divergences. These perspectives align with the message above: representer estimation and score/ratio learning can often be phrased as minimizing a feasible Bregman risk.

O. Losses, Link Functions, and Covariate Balancing

This appendix complements Section 6 by making explicit the balancing interpretation of ratio-based Riesz representers and by clarifying how the choice of (i) a loss (through the generator g in Theorem 6.1) and (ii) a link map (score-to-weight transformation) determines the resulting balancing weights. The discussion is aligned with generalized Riesz regression (Kato, 2026d).

O.1. Balancing Identities Implied by Ratio-Based Representers

Let q and p be two densities on $\mathcal{X}$ with $q \ll p$, and define the density ratio $r(x) := q(x)/p(x)$. Then, for any measurable test function $\varphi: \mathcal{X} \rightarrow \mathbb{R}$ such that the expectations exist,

$$\mathbb{E}_{X \sim p}[r(X)\varphi(X)] = \mathbb{E}_{X \sim q}[\varphi(X)]. \quad (19)$$

Thus, an exact ratio r is a moment-transport weight that maps expectations under p to those under q .

ATE balancing. Let $X = (D, Z)$ with $D \in \{1, -1\}$ and propensity $e_0(z) = \mathbb{P}(D = 1 \mid Z = z)$. The ATE representer is

$$\alpha_0^{\text{ATE}}(D, Z) = \frac{\mathbb{1}[D = 1]}{e_0(Z)} - \frac{\mathbb{1}[D = -1]}{1 - e_0(Z)}.$$

Then Eq. (19) implies that, for any $\varphi: \mathcal{Z} \rightarrow \mathbb{R}$,

$$\begin{aligned} \mathbb{E}\left[\mathbb{1}[D = 1]\frac{1}{e_0(Z)}\varphi(Z)\right] &= \mathbb{E}[\varphi(Z)], \\ \mathbb{E}\left[\mathbb{1}[D = -1]\frac{1}{1 - e_0(Z)}\varphi(Z)\right] &= \mathbb{E}[\varphi(Z)]. \end{aligned}$$

APE balancing. For APE under distribution shift, the representer is a difference of ratios. Let p_0 be the baseline density and p_1 and p_{-1} two counterfactual densities on $\mathcal{X}$ with $r_{\pm 1}(x) := p_{\pm 1}(x)/p_0(x)$. Then $\alpha_0^{\text{APE}}(x) = r_1(x) - r_{-1}(x)$ and, for any $\varphi: \mathcal{X} \rightarrow \mathbb{R}$,

$$\mathbb{E}_{X \sim p_0}[r_1(X)\varphi(X)] = \mathbb{E}_{X \sim p_1}[\varphi(X)], \quad \mathbb{E}_{X \sim p_0}[r_{-1}(X)\varphi(X)] = \mathbb{E}_{X \sim p_{-1}}[\varphi(X)].$$

The same identities apply to each fixed- δ point on the policy path, where $\alpha_{0,\delta} = r_{+\delta} - r_{-\delta}$.

O.2. Loss and Link Functions in ScoreMatchingRiesz

Link functions (mapping scores to weights). ScoreMatchingRiesz estimates log ratios (or log odds) by integrating score fields and then applies a link map.

- **Log link (ratios).** For endpoints (q, p) connected by a bridge $\{p_t\}_{t \in [0,1]}$, the endpoint log ratio satisfies $\log\{q(x)/p(x)\} = \int_1^0 \partial_t \log p_t(x) dt$ (Section 3). Time-ScoreMatchingRiesz therefore sets

$$\widehat{r}(x) := \exp(\widehat{\log r}(x)), \quad \widehat{\log r}(x) := \int_1^0 s_{\theta,t}^{\text{time}}(x) dt,$$

(optionally with clipping and mean-one calibration; Appendix M.2). The exponential map enforces $\widehat{r}(x) > 0$ by construction.

- **Logit link (propensities).** In ATE with $D \in \{1, -1\}$, write $\pi := \mathbb{P}(D = 1)$ and $p_{\pm 1}(z) = p_0(z \mid D = \pm 1)$. Then

$$\text{logit } e_0(z) = \log \frac{e_0(z)}{1 - e_0(z)} = \log \frac{\pi}{1 - \pi} + \log \frac{p_1(z)}{p_{-1}(z)}.$$

Using Time-ScoreMatchingRiesz to estimate $\log\{p_1(z)/p_{-1}(z)\}$ by score integration yields a logistic-link estimator

$$\widehat{e}(z) = \frac{1}{1 + \exp\left\{-\left(\log \frac{\pi}{1 - \pi} + \widehat{\log \frac{p_1}{p_{-1}}}(z)\right)\right\}}.$$

Loss functions (how the score model is trained). The role of the loss in Time-ScoreMatchingRiesz is played by the time-score fitting objective. Theorem 6.1 shows that the squared-loss objective is a special case of a Bregman family indexed by g . Changing g changes which local discrepancies in the score field are penalized more heavily and thus affects the implied weights after exponentiation.

Normalization as a balancing constraint. A minimal balancing condition for ratios is $\mathbb{E}_{X \sim p}[r(X)] = 1$, which is Eq. (19) with $\varphi \equiv 1$. Because ratios recovered from exponentiated integrals can be shifted by an additive constant in $\widehat{\log r}$, we impose cross-fitted mean-one calibration on the training fold and apply conservative log clipping before exponentiation when needed (Appendix M.2).

O.3. Automatic Covariate Balancing and Neyman Error

Automatic covariate balancing (generalized Riesz regression). Let $\phi : \mathcal{X} \rightarrow \mathbb{R}^p$ be a feature map and consider a representer model $\alpha_{\beta}(x)$ with a link matched to a Bregman generator g (e.g., exponential tilting for KL-type losses). For generalized Riesz regression (Appendix N), the first-order optimality/KKT conditions imply approximate sample balancing:

$$\left| \frac{1}{n} \sum_{i=1}^n \left\{ \alpha_{\widehat{\beta}}(X_i) \phi_j(X_i) - m(W_i, \phi_j) \right\} \right| \leq \lambda, \quad j = 1, \dots, p, \quad (20)$$

with equality under exact balance when $\lambda = 0$ (Kato, 2026d). When α_0 is a density ratio, Eq. (20) is a direct sample covariate-balancing statement.

Implication for the Neyman error. For the orthogonal estimator $\widehat{\theta} = \mathbb{E}_n[m(W, \widehat{\gamma}) + \widehat{\alpha}(X)\{Y - \widehat{\gamma}(X)\}]$, the leading remainder is driven by products of nuisance errors. A convenient sample diagnostic is the imbalance term $\mathbb{E}_n[\widehat{\alpha}(X)\widehat{\gamma}(X) - m(W, \widehat{\gamma})]$. If $\widehat{\gamma}$ is (approximately) in the span of $\{\phi_j\}$, then Eq. (20) controls this imbalance and reduces the Neyman remainder; in linear settings it can vanish under exact balance ($\lambda = 0$), yielding “automatic” orthogonalization (Kato, 2026d; Bruns-Smith et al., 2025).

Connection to augmented balancing weights. In linear settings, augmented balancing weights can be rewritten as a single regression estimator with coefficients that interpolate between a base regression and unregularized least squares, clarifying how weight regularization interacts with regression regularization (Bruns-Smith et al., 2025).

Table 4. Availability of APE/policy-path representer construction in the additional designs. Data-SMR refers to the data-score-based pushforward construction in Appendix G; Time-SMR refers to endpoint-sample-based ratio learning via time scores (DRE- ∞).

Design	Data-SMR	Time-SMR
Case 1: High-dimensional Gaussian P_0 , sparse outcome, shift policy	yes (pushforward)	yes
Case 2: High-dimensional Gaussian mixture P_0 , random-feature outcome, shift policy	yes (pushforward)	yes
Case 3: Stochastic intervention (resampling $D \mid Z$)	no (non-pushforward)	yes

P. Additional Synthetic Simulation Designs

In addition to the baseline Gaussian design in Section 9, we consider three additional synthetic designs (Cases 1–3). Cases 1–2 use shift policies (known pushforwards), so the APE/policy-path representer can be constructed from the data score of P_0 (Appendix G). Case 3 uses a stochastic intervention that changes $D \mid Z$ and is not a pushforward; in this case the data-score construction is unavailable for APE/policy-path estimation, and we rely on Time-SMR. In Case 3, we also compare Time-SMR with and without joint training.

Across all cases, we follow the evaluation protocol in Section 9: two-fold cross-fitting, the same MLP architecture for $\hat{\gamma}$, and trial-wise bias, MSE, and empirical 95% coverage, with ground truth approximated by population Monte Carlo. For shift policies, we also plot $\delta \mapsto \theta_0(\delta)$ on a grid $\delta \in [0, \delta_{\max}]$.

P.1. Case 1: High-Dimensional Gaussian Regressors with Sparse Outcome under a Shift Policy

This design increases the regressor dimension and makes the outcome regression sparse.

Regressors. Let $X = (D, Z) \in \mathbb{R}^p$ with $p = 50$. Draw

$$X \sim P_0 \equiv \mathcal{N}(0, \Sigma),$$

where Σ is Toeplitz (correlations decay with coordinate distance). We set $D = X^{(1)}$ and $Z = (X^{(2)}, \dots, X^{(p)})$.

Outcome. Generate

$$Y = \mu(X) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1),$$

where μ depends on D and only $k = 10$ coordinates of Z through nonlinear additive terms and interactions. The remaining coordinates enter only through a small nuisance component.

Policy and targets. We consider the deterministic shift

$$T_\delta(d, z) = (d + \delta, z), \quad X_\delta = T_\delta(X), \quad X \sim P_0,$$

and the symmetric policy path

$$\theta_0(\delta) = \mathbb{E}[\mu(D + \delta, Z)] - \mathbb{E}[\mu(D - \delta, Z)], \quad \delta \in [0, \delta_{\max}].$$

Because $P_\delta = (T_\delta)_\# P_0$ is a known pushforward, the APE/path ratios can be constructed from the observational data score (Appendix G). Time-SMR is also applicable using endpoint samples obtained by shifting D .

Table 5 shows that Data-SMR yields the smallest AME MSE, while Riesz regression is highly unstable. For APE, Time-SMR attains a much smaller MSE than Data-SMR but exhibits severe undercoverage, suggesting overly narrow confidence intervals. Data-SMR has larger bias/MSE for APE but substantially closer-to-nominal coverage.

P.2. Case 2: High-Dimensional Gaussian Mixture Regressors with Random-Feature Outcomes under a Shift Policy

This design makes P_0 multimodal and uses a dense outcome regression.

Regressors. Let $X = (D, Z) \in \mathbb{R}^p$ with $p = 50$. Draw

$$X \sim P_0 \equiv \frac{1}{K} \sum_{k=1}^K \mathcal{N}(\mu^{(k)}, \Sigma),$$

with shared Toeplitz covariance Σ and separated means $\mu^{(k)}$. We use $K = 4$.

Table 5. Case 1 results.

AME (true = 2.214)			
Metric	Data-SMR	Time-SMR	Riesz reg.
Bias	0.299	-0.198	22.828
MSE	3.272	10.736	2015.537
Cov. (95%)	88.0%	84.7%	79.2%

APE (true = 4.042)			
Metric	Data-SMR	Time-SMR	Riesz reg.
Bias	3.465	-1.162	27.828
MSE	133.853	3.872	3066.051
Cov. (95%)	85.6%	16.2%	77.7%

Table 6. Case 2 results.

AME (true = 0.396)			
Metric	Data-SMR	Time-SMR	Riesz reg.
Bias	0.128	0.187	1.424
MSE	0.092	0.715	106.067
Cov. (95%)	83.9%	86.7%	83.5%

APE (true = 0.843)			
Metric	Data-SMR	Time-SMR	Riesz reg.
Bias	0.206	0.212	0.669
MSE	0.687	0.097	128.748
Cov. (95%)	86.2%	20.1%	82.9%

Outcome. Generate

$$Y = \mu(X) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1),$$

where μ includes low-order terms in D and a dense random-feature component $x \mapsto v^\top \tanh(Wx + b)$ (200 random features in our experiments), plus interactions.

Policy and targets. We use the same shift $T_\delta(d, z) = (d + \delta, z)$ and the same AME/APE targets as in Case 1. Data-SMR is applicable for APE via the pushforward construction, and Time-SMR is applicable via endpoint samples.

Table 6 shows a similar pattern: Data-SMR performs best for AME, while Riesz regression is unstable. For APE, Time-SMR has the smallest MSE but exhibits severe undercoverage. Data-SMR has larger MSE but markedly better coverage.

P.3. Case 3: Non-Pushforward Stochastic Policies via Resampling $D \mid Z$

This design uses stochastic interventions that are not known pushforwards of P_0 . As a result, the APE/path representer cannot be constructed from the observational data score alone.

Observational law and outcome. Use the baseline Gaussian regressor design from Section 9:

$$X = (D, Z) \sim \mathcal{N}(0, \Sigma), \quad D = X^{(1)}, \quad Z = (X^{(2)}, X^{(3)}),$$

and generate $Y = \mu(X) + \varepsilon$ with the same μ and $\varepsilon \sim \mathcal{N}(0, 1)$.

Stochastic policies. Fix $\delta_{\max} > 0$. For each $\delta \in [0, \delta_{\max}]$, define $P_{+\delta}$ and $P_{-\delta}$ by drawing $Z \sim p_0(z)$ and resampling

$$D^{(\pm, \delta)} \mid Z = z \sim \mathcal{N}(m_{\pm, \delta}(z), s_{\pm}(z)^2), \quad X^{(\pm, \delta)} = (D^{(\pm, \delta)}, Z).$$

Table 7. Case 3 results.

AME (true = 2.225)		
Metric	Time-SMR (without joint training)	Time-SMR (with joint training)
Bias	0.012	0.012
MSE	0.147	0.147
Cov. (95%)	86.0%	86.0%
APE (true = 1.324)		
Metric	Time-SMR (without joint training)	Time-SMR (with joint training)
Bias	-0.949	-0.511
MSE	7.392	1.254
Cov. (95%)	82.0%	74.0%

We set

$$m_{\pm,\delta}(z) = \mathbb{E}_{P_0}[D \mid Z = z] \pm \delta b(z), \quad b(z) = \tanh(z^{(1)}) + \frac{1}{2} \sin(z^{(2)}),$$

and

$$s_{\pm}(z) = \sqrt{\text{Var}_{P_0}(D \mid Z = z)} \left(0.8 + 0.4 \sigma(\pm z^{(1)}) \right), \quad \sigma(u) = \frac{1}{1 + e^{-u}}.$$

Then $p_{\pm,\delta}(d, z) = p_0(z) p_{\pm,\delta}(d \mid z)$.

Non-pushforward implication. Because the policy changes the conditional law of $D \mid Z$, there is generally no deterministic map T_{δ} such that $X^{(\pm,\delta)} = T_{\delta}(X)$ with $X \sim P_0$. Thus the ratio $p_{\pm,\delta}/p_0$ is not determined by p_0 (or its data score) alone, and the data-score-based APE construction is unavailable. Time-SMR applies by treating $(q, p) = (p_{\pm,\delta}, p_0)$ and using simulated samples from $p_{\pm,\delta}$ together with observed samples from p_0 .

APE and results. Define

$$\theta_0(\delta) = \mathbb{E}_{X \sim p_{+,\delta}}[\gamma_0(X)] - \mathbb{E}_{X \sim p_{-,\delta}}[\gamma_0(X)], \quad \delta \in [0, \delta_{\max}],$$

and approximate the truth by Monte Carlo draws from $p_{+,\delta}$ and $p_{-,\delta}$.

Table 7 illustrates the non-pushforward regime, where the data-score-based APE/policy-path construction is unavailable. AME performance is essentially unchanged by joint training. For APE, joint training reduces bias and MSE, but coverage remains below nominal for both variants (74–82%).

Q. Empirical Application: Policy Paths of Monetary Policy Shifts on Equity Returns

This appendix describes an empirical application of ScoreMatchingRiesz using publicly available U.S. financial and macroeconomic data². The empirical objective is *methodological*: we demonstrate how a single score-based Riesz pipeline produces (i) an AME, (ii) a finite-shift APE, and (iii) an entire *policy path* $\delta \mapsto \theta_h(\delta)$, and we compare these outputs to a standard *Riesz regression* baseline in terms of point estimates, confidence intervals, and policy-path visualizations. No “true” effects are assumed or required. Because the data form a dependent time series, the i.i.d. theory in the main text does not directly cover this setting, and we interpret the results as a methodological demonstration. We do not rely on train–test splits; all estimation uses the full sample.

Q.1. What We Estimate and Why: Local Projections Meet Policy Paths

Empirical work in macro-finance often seeks to quantify how an observable policy variable (e.g. an interest rate move) is associated with future asset returns. A standard approach is the *local projection* (LP) method (Jordà, 2005), which estimates horizon-specific responses by regressing future outcomes on current policy moves and controls. In the simplest

²We also conducted empirical analysis on Japanese monetary policy in our subsequent work (Kato, 2026e).

linear LP, one estimates a coefficient β_h from

$$Y_{t,h} = \beta_h D_t + \text{controls} + \text{error},$$

where $Y_{t,h}$ is a forward outcome over horizon h , and D_t is a policy move at time t .

Our goal is to move beyond a linear coefficient and estimate a *nonlinear, policy-shift response curve* that maps the size of a policy shift δ to the implied change in outcomes. This curve is the *policy path*:

$$\delta \mapsto \theta_h(\delta),$$

where $\theta_h(\delta)$ is a symmetric effect of shifting the policy move by $+\delta$ versus $-\delta$. The path nests both local and global objects:

- The **AME** corresponds to the *local slope* of the path at the origin.
- The **APE** corresponds to the path evaluated at a particular policy shift δ_{policy} (e.g. 25 bps).

This design makes it natural to compare different Riesz-representer estimators: ScoreMatchingRiesz constructs the path by estimating an AME score and integrating it to obtain density ratios, while Riesz regression estimates the representer directly by minimizing a Riesz loss.

Q.2. Data Sources and Preprocessing

Data sources (replicable, public). We combine: (i) macroeconomic, monetary, and financial indicators from the Federal Reserve Economic Data (FRED) database, and (ii) asset prices from Yahoo Finance (downloaded via `yfinance`). We choose these sources to ensure replicability with a short script.

Sampling frequency and alignment. All series are converted to monthly frequency and aligned on end-of-month dates. Daily asset prices are converted to month-end adjusted close prices and then to monthly log returns. Daily macro-financial series (e.g. volatility indices) are aggregated to monthly averages. We merge all series on the monthly index and drop months with missing values after constructing lags and forward outcomes.

Variables. Let $t = 1, \dots, T$ index months. We define:

- **Policy variable.** D_t is the *monthly change* in an interest-rate proxy, measured in percentage points. A baseline choice is

$$D_t := \Delta \text{FF}_t,$$

where FF_t is an effective federal funds rate series. We use changes rather than levels to reduce persistence and to align with a “policy-move” interpretation.

- **Market outcome.** Let P_t denote the end-of-month adjusted close price of an equity index proxy (e.g. an S&P 500 ETF). Define the one-month log return $r_t := \log P_t - \log P_{t-1}$. We construct horizon- h forward outcomes as cumulative returns:

$$Y_{t,h} := \sum_{j=1}^h r_{t+j}, \quad h \in \{1, \dots, H\}. \quad (21)$$

Optionally, one can use excess returns $r_t^e = r_t - r_t^f$ by subtracting a short-rate proxy.

- **Controls and lag structure.** Z_t collects macro-financial controls and lags, in the spirit of LP specifications:

$$Z_t = (\text{macro controls at } t, \text{ lags of } (D, Z)).$$

The macro control set can include inflation, unemployment, industrial production growth, term spreads, and volatility proxies. We include p lags of key controls and D_t (e.g. $p = 12$ months) to capture state dependence and dynamics.

- **Factor controls (PCA).** To summarize common movements in a panel of asset returns, we extract a small number of latent factors by PCA. Let $R_t \in \mathbb{R}^M$ be a panel of monthly log returns for a set of liquid ETFs (e.g. sector ETFs). After standardizing each return series, we compute the first K principal components:

$$F_t = \text{PCA}_K(R_t) \in \mathbb{R}^K,$$

and append F_t (and optionally its lags) to Z_t . This introduces a low-dimensional factor structure consistent with asset-pricing intuition while keeping the nuisance learning flexible.

Regressor vector. We define the regressor

$$X_t := (D_t, Z_t) \in \mathbb{R}^d,$$

and the horizon- h “observation” as $W_{t,h} := (X_t, Y_{t,h})$.

Q.3. Targets and Their Riesz Representers: AME, APE, and Policy Paths

For each horizon h , define the conditional mean (LP regression function)

$$\gamma_{0,h}(x) := \mathbb{E}[Y_{t,h} \mid X_t = x], \quad x \in \mathbb{R}^d.$$

We study three related parameters.

Q.3.1. AME AS A LINEAR FUNCTIONAL AND ITS REPRESENTER

The horizon- h average marginal effect (AME) is

$$\theta_{0,h}^{\text{AME}} := \mathbb{E}[\partial_d \gamma_{0,h}(D_t, Z_t)],$$

where ∂_d denotes the partial derivative with respect to the policy coordinate d . Under standard integration-by-parts conditions (e.g. sufficient smoothness and boundary decay), this functional admits a Riesz representer $\alpha_0^{\text{AME}} \in L_2(P_X)$ satisfying

$$\mathbb{E}[\partial_d \gamma(D, Z)] = \mathbb{E}[\alpha_0^{\text{AME}}(D, Z) \gamma(D, Z)] \quad \text{for all } \gamma \in L_2(P_X),$$

with the explicit form

$$\alpha_0^{\text{AME}}(D, Z) = -\partial_d \log p_0(D, Z), \tag{22}$$

where p_0 is the joint density of $X = (D, Z)$ under the stationary distribution of the observed process. This identity motivates score-based estimation of the AME representer.

Q.3.2. POLICY SHIFTS, APE, AND THE POLICY PATH

We consider *smooth policy shifts* that translate the policy coordinate:

$$\tau_\delta(d, z) := (d + \delta, z), \quad \delta \in \mathbb{R}.$$

Let p_δ be the distribution of $X_\delta = \tau_\delta(X)$ when $X \sim p_0$. Then $p_\delta(d, z) = p_0(d - \delta, z)$ (ignoring boundary effects).

Policy path. For horizon h , define the symmetric policy-path parameter

$$\theta_{0,h}(\delta) := \mathbb{E}_{X \sim p_\delta}[\gamma_{0,h}(X)] - \mathbb{E}_{X \sim p_0}[\gamma_{0,h}(X)]. \tag{23}$$

By a change of variables, Eq. (23) is equivalently written as an expectation under p_0 :

$$\theta_{0,h}(\delta) = \mathbb{E}[\gamma_{0,h}(D_t + \delta, Z_t) - \gamma_{0,h}(D_t - \delta, Z_t)]. \tag{24}$$

This shows that the policy path is a *linear functional* of $\gamma_{0,h}$.

APE as a point on the path. Fix a policy magnitude of interest $\delta_{\text{policy}} > 0$ (e.g. 0.25 percentage points, i.e. 25 bps). We define the finite-shift APE at horizon h as

$$\theta_{0,h}^{\text{APE}} := \theta_{0,h}(\delta_{\text{policy}}). \tag{25}$$

Path-AME link (local approximation). Differentiating Eq. (24) at $\delta = 0$ yields

$$\theta'_{0,h}(0) = \mathbb{E}[\partial_d \gamma_{0,h}(D, Z)] + \mathbb{E}[\partial_d \gamma_{0,h}(D, Z)] = 2\theta_{0,h}^{\text{AME}}.$$

Hence

$$\theta_{0,h}^{\text{AME}} = \frac{1}{2}\theta'_{0,h}(0), \quad \theta_{0,h}(\delta) \approx 2\delta \cdot \theta_{0,h}^{\text{AME}} \text{ for small } \delta. \quad (26)$$

This motivates visual checks that compare the estimated path to the local linear approximation $2\delta\hat{\theta}_h^{\text{AME}}$ near the origin.

Q.3.3. RIESZ REPRESENTER FOR THE POLICY PATH AND ITS DENSITY-RATIO FORM

Define the moment map for the policy path:

$$m_\delta(W_{t,h}, \gamma) := \gamma(D_t + \delta, Z_t) - \gamma(D_t - \delta, Z_t). \quad (27)$$

This depends only on X_t and is linear in γ . By the Riesz representation theorem, there exists $\alpha_{0,\delta} \in L_2(P_X)$ such that

$$\mathbb{E}[m_\delta(W_{t,h}, \gamma)] = \mathbb{E}[\alpha_{0,\delta}(X_t)\gamma(X_t)] \quad \text{for all } \gamma \in L_2(P_X). \quad (28)$$

Importantly, because m_δ depends only on X_t , the representer $\alpha_{0,\delta}$ is *common across horizons* h . Thus, in both of our approaches below, representer estimation is done once (per δ) and reused across all horizons.

Density-ratio representation under translation shifts. Define the importance weight

$$r_\delta(x) := \frac{p_\delta(x)}{p_0(x)}.$$

Then

$$\mathbb{E}_{X \sim p_\delta}[\gamma(X)] = \mathbb{E}_{X \sim p_0}[r_\delta(X)\gamma(X)],$$

and therefore Eq. (23) implies

$$\alpha_{0,\delta}(x) = r_\delta(x) - r_{-\delta}(x).$$

Under $p_\delta(d, z) = p_0(d - \delta, z)$, this ratio has the explicit form

$$r_\delta(d, z) = \frac{p_0(d - \delta, z)}{p_0(d, z)}.$$

From AME scores to ratios (the key ScoreMatchingRiesz identity). By the fundamental theorem of calculus,

$$\begin{aligned} \log r_\delta(d, z) &= \log p_0(d - \delta, z) - \log p_0(d, z) \\ &= - \int_0^\delta \partial_u \log p_0(d - u, z) du. \end{aligned} \quad (29)$$

Combining Eq. (22) and Eq. (29) shows that estimating the AME score $\partial_d \log p_0(d, z)$ enables construction of *density ratios for all* δ by one-dimensional numerical integration, and hence yields the entire policy path.

Q.4. Estimation I: ScoreMatchingRiesz for AME, APE, and the Policy Path

We now describe the proposed estimator, emphasizing the “one score model $\Rightarrow$ all δ ” reuse.

Q.4.1. STEP 1: ESTIMATE THE TREATMENT SCORE BY DATA-SCOREMATCHINGRIESZ

We estimate the treatment-direction score $\partial_d \log p_0(d, z)$ via Data-ScoreMatchingRiesz, which follows denoising score matching (DSM), injecting Gaussian noise only in the policy coordinate. Let $\varepsilon \sim \mathcal{N}(0, 1)$ be independent and draw $\tau \sim \text{Unif}[0, 1]$. Define

$$\tilde{D} = D + \sigma(\tau)\varepsilon,$$

where $\sigma(\tau)$ is a decreasing noise schedule (large noise at small τ , small noise at τ near 1). We fit a neural model $s_\theta(\tilde{d}, z, \tau)$ minimizing

$$\mathcal{R}^{\text{DSM}}(\theta) = \mathbb{E} \left[\lambda(\tau) \left(s_\theta(D + \sigma(\tau)\varepsilon, Z, \tau) + \frac{\varepsilon}{\sigma(\tau)} \right)^2 \right].$$

For each fixed τ , the population minimizer equals the score of the $\sigma(\tau)$ -smoothed density in the $\tilde{d}$ direction. We then evaluate at a small-noise time $\tau_{\text{eval}} \approx 1$ to approximate the unsmoothed score and define

$$\hat{\alpha}^{\text{AME}}(D, Z) := -s_{\hat{\theta}}(D, Z, \tau_{\text{eval}}).$$

Because this step uses only the regressor process $\{X_t\}$, it is performed once and reused for all horizons h .

Q.4.2. STEP 2: CONSTRUCT RATIOS AND PATH REPRESENTERS FOR A GRID OF POLICY SHIFTS

Fix a grid $\mathcal{D} = \{\delta_1, \dots, \delta_M\}$ of nonnegative policy shifts (e.g. from 0 to 1 percentage point). For each $\delta \in \mathcal{D}$, we estimate the log ratio using Eq. (29) with the learned score:

$$\widehat{\log r_\delta}(d, z) := - \int_0^\delta s_{\hat{\theta}}(d - u, z, \tau_{\text{eval}}) du, \quad \hat{r}_\delta(d, z) = \exp(\widehat{\log r_\delta}(d, z)). \quad (30)$$

The policy-path representer is then

$$\hat{\alpha}_\delta^{\text{path}}(x) := \hat{r}_\delta(x) - \hat{r}_{-\delta}(x). \quad (31)$$

In implementation, the integral in Eq. (30) is computed by a trapezoidal rule on a fine grid in u . To stabilize heavy tails, we optionally clip $\widehat{\log r_\delta}$ before exponentiation and/or clip $\hat{\alpha}_\delta^{\text{path}}$.

Q.4.3. STEP 3: HORIZON-SPECIFIC REGRESSION AND ORTHOGONAL SCORES

For each horizon h , we estimate $\gamma_{0,h}$ by a flexible regressor $\hat{\gamma}_h$ trained on the full sample $\{(X_t, Y_{t,h})\}_{t=1}^T$. Denote residuals $\hat{U}_{t,h} := Y_{t,h} - \hat{\gamma}_h(X_t)$.

AME (direct). We estimate $\theta_{0,h}^{\text{AME}}$ by the orthogonal-score estimator

$$\hat{\theta}_{h,\text{SMR}}^{\text{AME}} = \frac{1}{T} \sum_{t=1}^T \left\{ \partial_d \hat{\gamma}_h(X_t) + \hat{\alpha}^{\text{AME}}(X_t) \hat{U}_{t,h} \right\}. \quad (32)$$

The derivative $\partial_d \hat{\gamma}_h(X_t)$ is computed by automatic differentiation.

Policy path and APE. For each $\delta \in \mathcal{D}$, we estimate the policy path by

$$\hat{\theta}_{h,\text{SMR}}(\delta) = \frac{1}{T} \sum_{t=1}^T \left[\hat{\gamma}_h(D_t + \delta, Z_t) - \hat{\gamma}_h(D_t - \delta, Z_t) + \hat{\alpha}_\delta^{\text{path}}(X_t) \hat{U}_{t,h} \right]. \quad (33)$$

The APE is $\hat{\theta}_{h,\text{SMR}}^{\text{APE}} = \hat{\theta}_{h,\text{SMR}}(\delta_{\text{policy}})$. Equation Eq. (33) yields the entire curve $\delta \mapsto \hat{\theta}_{h,\text{SMR}}(\delta)$ in one pass once ratios are available.

Q.5. Estimation II: Riesz Regression Baseline for AME, APE, and the Policy Path

We compare ScoreMatchingRiesz to a Riesz-regression baseline that estimates the policy-path representer *directly* from the defining Riesz property Eq. (28), without imposing a density-ratio structure.

Q.5.1. RIESZ REGRESSION OBJECTIVE FOR THE POLICY-PATH REPRESENTER

For a fixed δ , define the functional m_δ as in Eq. (27). The Riesz representer $\alpha_{0,\delta}$ is the unique minimizer (in $L_2(P_X)$) of the population quadratic objective

$$\mathcal{L}_\delta(\alpha) = \mathbb{E}[\alpha(X)^2] - 2\mathbb{E}[m_\delta(W, \alpha)] + \text{const.},$$

where the additive constant does not depend on α . The corresponding sample objective (with ridge regularization) is

$$\hat{\alpha}_{\delta, \text{RR}} \in \arg \min_{\alpha \in \mathcal{A}} \left\{ \mathbb{E}_T[\alpha(X_t)^2] - 2\mathbb{E}_T[\alpha(D_t + \delta, Z_t) - \alpha(D_t - \delta, Z_t)] + \lambda \|\alpha\|_{\mathcal{A}}^2 \right\}, \quad (34)$$

where $\mathbb{E}_T[\cdot] = T^{-1} \sum_{t=1}^T (\cdot)$, $\mathcal{A}$ is a function class (e.g. a sieve or a neural net), and $\lambda > 0$ is a regularization parameter.

Sieve implementation (closed form). In our implementation we use a linear sieve $\alpha(x) = \beta^\top \phi(x)$ with feature map $\phi(x) \in \mathbb{R}^p$ (e.g. linear or low-degree polynomial features, or random features). Plugging into Eq. (34) yields the ridge-regularized quadratic problem with closed-form solution

$$\hat{\beta}_\delta = \left(\hat{\Sigma} + \lambda I \right)^{-1} \hat{b}_\delta, \quad \hat{\Sigma} = \mathbb{E}_T[\phi(X_t)\phi(X_t)^\top], \quad \hat{b}_\delta = \mathbb{E}_T[\phi(D_t + \delta, Z_t) - \phi(D_t - \delta, Z_t)].$$

We then set $\hat{\alpha}_{\delta, \text{RR}}^{\text{path}}(x) = \hat{\beta}_\delta^\top \phi(x)$. As with the score-based method, this representer depends only on $\{X_t\}$ and is common across horizons h .

Q.5.2. ORTHOGONAL POLICY-PATH ESTIMATOR USING THE RIESZ REGRESSION REPRESENTER

Given the same horizon-specific regression $\hat{\gamma}_h$ and residuals $\hat{U}_{t,h}$, we estimate the policy path by

$$\hat{\theta}_{h, \text{RR}}(\delta) = \frac{1}{T} \sum_{t=1}^T \left[\hat{\gamma}_h(D_t + \delta, Z_t) - \hat{\gamma}_h(D_t - \delta, Z_t) + \hat{\alpha}_{\delta, \text{RR}}^{\text{path}}(X_t) \hat{U}_{t,h} \right], \quad \delta \in \mathcal{D}. \quad (35)$$

The APE baseline is $\hat{\theta}_{h, \text{RR}}^{\text{APE}} = \hat{\theta}_{h, \text{RR}}(\delta_{\text{policy}})$.

AME from the path (baseline and cross-check). Because $\theta_{0,h}^{\text{AME}} = \frac{1}{2} \theta'_{0,h}(0)$ by Eq. (26), we compute a path-based AME estimate for the Riesz regression baseline using a small δ_{AME} :

$$\hat{\theta}_{h, \text{RR}}^{\text{AME}} := \frac{1}{2\delta_{\text{AME}}} \hat{\theta}_{h, \text{RR}}(\delta_{\text{AME}}). \quad (36)$$

We report Eq. (36) alongside the direct AME estimate Eq. (32) from ScoreMatchingRiesz, and we also check the local linear approximation $2\delta \hat{\theta}_h^{\text{AME}}$ against the estimated paths near $\delta = 0$.

Q.6. Inference for Time-Series Data: Pointwise HAC Confidence Bands

Because $\{(X_t, Y_{t,h})\}$ is a time series, we compute heteroskedasticity-and-autocorrelation consistent (HAC) standard errors for both methods, treating the orthogonal-score summands as estimated influence functions.

Estimated influence functions. Define the centered summands for the policy path:

$$\hat{\psi}_{t,h, \text{SMR}}(\delta) := \hat{\gamma}_h(D_t + \delta, Z_t) - \hat{\gamma}_h(D_t - \delta, Z_t) + \hat{\alpha}_\delta^{\text{path}}(X_t) \hat{U}_{t,h} - \hat{\theta}_{h, \text{SMR}}(\delta), \quad \hat{\psi}_{t,h, \text{RR}}(\delta) := \hat{\gamma}_h(D_t + \delta, Z_t) - \hat{\gamma}_h(D_t - \delta, Z_t) + \hat{\alpha}_{\delta, \text{RR}}^{\text{path}}(X_t) \hat{U}_{t,h} - \hat{\theta}_{h, \text{RR}}(\delta)$$

These sequences have mean approximately zero by construction. We estimate the variance of $T^{-1} \sum_t \hat{\psi}_{t,h}(\delta)$ by a Newey–West estimator:

$$\hat{\Omega}_h(\delta) = \hat{\Gamma}_0(\delta) + 2 \sum_{\ell=1}^{L_h} w_\ell \hat{\Gamma}_\ell(\delta), \quad \hat{\Gamma}_\ell(\delta) = \frac{1}{T} \sum_{t=\ell+1}^T \hat{\psi}_{t,h}(\delta) \hat{\psi}_{t-\ell,h}(\delta), \quad w_\ell = 1 - \frac{\ell}{L_h + 1}, \quad (37)$$

and set the standard error to $\widehat{\text{se}}_h(\delta) = \sqrt{\hat{\Omega}_h(\delta)/T}$. We use a horizon-dependent lag truncation L_h (e.g. $L_h = \max\{6, h\}$ for monthly data) and report pointwise 95% intervals

$$\hat{\theta}_h(\delta) \pm 1.96 \widehat{\text{se}}_h(\delta),$$

computed separately for ScoreMatchingRiesz and for Riesz regression. These are *pointwise* bands in δ (no multiple-testing adjustment).

Figure 3. Illustration of the policy path $\delta \mapsto \hat{\theta}_h(\delta)$ at a fixed horizon h . The solid lines show point estimates from ScoreMatchingRiesz and from the Riesz regression baseline. Shaded areas are pointwise HAC 95% confidence bands. A dotted line may show the local approximation $2\delta\hat{\theta}_h^{\text{AME}}$ near $\delta = 0$.

Remarks on nuisance estimation. We do not split the sample into train and test subsets. All nuisance functions are trained on the full sample. Because the i.i.d. assumptions underlying the main theoretical results do not directly apply to dependent time-series data, we view the HAC bands as a pragmatic approximation commonly used in empirical macro-finance when first-stage learners are regularized, and we use them primarily for comparing representer estimators. As a robustness check, one can implement block cross-fitting by splitting the time index into contiguous folds and evaluating the orthogonal scores on held-out blocks.

Q.7. Reporting and Visualization: Comparing Estimators and Confidence Bands

Our empirical outputs consist of:

Numerical comparison (AME and APE). For each horizon h , we report a table containing:

- $\hat{\theta}_{h,\text{SMR}}^{\text{AME}}$ from Eq. (32) with HAC 95% CI.
- $\hat{\theta}_{h,\text{RR}}^{\text{AME}}$ from Eq. (36) with HAC 95% CI (using the HAC CI for $\hat{\theta}_{h,\text{RR}}(\delta_{\text{AME}})$ rescaled by $1/(2\delta_{\text{AME}})$).
- $\hat{\theta}_{h,\text{SMR}}^{\text{APE}} = \hat{\theta}_{h,\text{SMR}}(\delta_{\text{policy}})$ with HAC 95% CI.
- $\hat{\theta}_{h,\text{RR}}^{\text{APE}} = \hat{\theta}_{h,\text{RR}}(\delta_{\text{policy}})$ with HAC 95% CI.

This provides a direct comparison of point estimates and uncertainty at economically meaningful policy magnitudes.

Policy-path plots. For selected horizons (e.g. $h \in \{1, 3, 6, 12\}$ months), we plot the entire policy path for both methods:

- The ScoreMatchingRiesz path $\delta \mapsto \hat{\theta}_{h,\text{SMR}}(\delta)$ with its pointwise 95% band.
- The Riesz regression path $\delta \mapsto \hat{\theta}_{h,\text{RR}}(\delta)$ with its pointwise 95% band.

Optionally, we overlay the local linear approximation $2\delta\hat{\theta}_{h,\text{SMR}}^{\text{AME}}$ to highlight the degree of nonlinearity and the coherence between AME and the global path.

Interpretation of differences. The two approaches differ only in *how* the representer is estimated:

- ScoreMatchingRiesz estimates the AME score once and constructs $\hat{\alpha}_{\delta}^{\text{path}}$ for all δ via the integral identity Eq. (29). This enforces a density-ratio structure across δ and couples the entire path through a single learned score.
- Riesz regression estimates $\hat{\alpha}_{\delta,\text{RR}}^{\text{path}}$ directly by the Riesz loss Eq. (34), separately for each δ (or via a closed-form sieve), without imposing the density-ratio structure.

Comparing the resulting paths and confidence bands helps diagnose whether imposing the score-to-ratio structure stabilizes estimation, particularly for larger policy shifts where representers can become heavy-tailed.

Q.8. Implementation Details for Table 8

Table 8 reports the estimated *one-sided* policy-shift path

$$\theta_{0,h}(\delta, 0) := \mathbb{E} [\gamma_{0,h}(D_t + \delta, Z_t) - \gamma_{0,h}(D_t, Z_t)],$$

evaluated on a grid of shifts δ (in percentage points) and horizons $h \in \{1, \dots, 12\}$ months. This is the object produced by the estimation routine in our implementation. The table is organized in wide format with *rows grouped by δ* and *three subrows per δ* : the point estimate $\hat{\theta}_h(\delta, 0)$, its HAC standard error, and the pointwise 95% interval.

Table 8. Policy path table (wide): rows are grouped by δ with three subrows (theta, se, CI).

δ		$h = 1$	$h = 2$	$h = 3$	$h = 4$	$h = 5$	$h = 6$	$h = 7$	$h = 8$	$h = 9$	$h = 10$	$h = 11$	$h = 12$
-1.00	theta	-0.004	0.041	0.099	-0.009	-0.045	-0.064	-0.091	-0.112	-0.074	-0.016	-0.041	-0.064
	se	0.000	0.002	0.003	0.003	0.006	0.003	0.004	0.003	0.003	0.005	0.006	0.006
	[CI _{low} , CI _{high}]	[-0.005, -0.004]	[0.037, 0.044]	[0.092, 0.106]	[-0.015, -0.002]	[-0.058, -0.033]	[-0.071, -0.057]	[-0.099, -0.083]	[-0.118, -0.107]	[-0.079, -0.069]	[-0.027, -0.006]	[-0.053, -0.029]	[-0.076, -0.053]
-0.75	theta	-0.003	0.029	0.074	-0.007	-0.033	-0.056	-0.074	-0.088	-0.059	-0.016	-0.037	-0.054
	se	0.000	0.001	0.003	0.003	0.005	0.003	0.003	0.002	0.002	0.004	0.005	0.004
	[CI _{low} , CI _{high}]	[-0.003, -0.002]	[0.027, 0.032]	[0.069, 0.080]	[-0.012, -0.002]	[-0.042, -0.023]	[-0.061, -0.050]	[-0.080, -0.068]	[-0.093, -0.084]	[-0.063, -0.055]	[-0.024, -0.009]	[-0.046, -0.027]	[-0.063, -0.046]
-0.50	theta	-0.001	0.018	0.049	-0.005	-0.021	-0.043	-0.053	-0.062	-0.041	-0.014	-0.029	-0.040
	se	0.000	0.001	0.002	0.002	0.003	0.002	0.002	0.002	0.001	0.002	0.003	0.003
	[CI _{low} , CI _{high}]	[-0.002, -0.001]	[0.017, 0.020]	[0.046, 0.053]	[-0.009, -0.002]	[-0.028, -0.014]	[-0.047, -0.038]	[-0.057, -0.049]	[-0.065, -0.059]	[-0.044, -0.038]	[-0.019, -0.009]	[-0.035, -0.022]	[-0.046, -0.034]
-0.25	theta	-0.000	0.009	0.024	-0.003	-0.010	-0.025	-0.028	-0.032	-0.021	-0.008	-0.017	-0.022
	se	0.000	0.000	0.001	0.001	0.002	0.002	0.001	0.001	0.001	0.001	0.002	0.002
	[CI _{low} , CI _{high}]	[-0.001, -0.000]	[0.008, 0.010]	[0.023, 0.026]	[-0.004, -0.001]	[-0.014, -0.006]	[-0.028, -0.022]	[-0.031, -0.026]	[-0.034, -0.031]	[-0.023, -0.020]	[-0.011, -0.006]	[-0.021, -0.013]	[-0.025, -0.019]
0.00	theta	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	se	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	[CI _{low} , CI _{high}]	[0.000, 0.000]	[0.000, 0.000]	[0.000, 0.000]	[0.000, 0.000]	[0.000, 0.000]	[0.000, 0.000]	[0.000, 0.000]	[0.000, 0.000]	[0.000, 0.000]	[0.000, 0.000]	[0.000, 0.000]	[0.000, 0.000]
0.25	theta	-0.000	-0.008	-0.024	0.004	0.007	0.024	0.031	0.034	0.024	0.012	0.017	0.026
	se	0.000	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.002	0.002
	[CI _{low} , CI _{high}]	[-0.000, 0.000]	[-0.009, -0.007]	[-0.026, -0.022]	[0.003, 0.006]	[0.005, 0.009]	[0.021, 0.026]	[0.029, 0.033]	[0.032, 0.036]	[0.022, 0.025]	[0.009, 0.015]	[0.014, 0.020]	[0.022, 0.029]
0.50	theta	-0.000	-0.015	-0.046	0.009	0.014	0.053	0.066	0.070	0.048	0.026	0.039	0.054
	se	0.000	0.001	0.002	0.002	0.003	0.002	0.002	0.002	0.002	0.002	0.003	0.003
	[CI _{low} , CI _{high}]	[-0.001, -0.000]	[-0.017, -0.013]	[-0.050, -0.042]	[0.005, 0.012]	[0.008, 0.019]	[0.049, 0.057]	[0.062, 0.069]	[0.066, 0.073]	[0.045, 0.051]	[0.021, 0.031]	[0.032, 0.045]	[0.047, 0.060]
0.75	theta	-0.001	-0.021	-0.067	0.013	0.019	0.084	0.101	0.106	0.073	0.041	0.063	0.084
	se	0.000	0.001	0.003	0.003	0.004	0.003	0.003	0.002	0.002	0.003	0.005	0.005
	[CI _{low} , CI _{high}]	[-0.001, -0.000]	[-0.023, -0.018]	[-0.072, -0.061]	[0.008, 0.018]	[0.011, 0.027]	[0.078, 0.089]	[0.095, 0.107]	[0.101, 0.110]	[0.068, 0.077]	[0.034, 0.047]	[0.053, 0.073]	[0.074, 0.093]
1.00	theta	-0.001	-0.025	-0.086	0.018	0.023	0.116	0.137	0.142	0.098	0.057	0.090	0.115
	se	0.000	0.002	0.004	0.004	0.005	0.004	0.004	0.003	0.003	0.004	0.007	0.007
	[CI _{low} , CI _{high}]	[-0.002, -0.001]	[-0.028, -0.022]	[-0.093, -0.079]	[0.011, 0.025]	[0.013, 0.034]	[0.109, 0.123]	[0.130, 0.145]	[0.136, 0.148]	[0.092, 0.104]	[0.048, 0.065]	[0.077, 0.102]	[0.102, 0.128]

Shift grid and horizons. We use a symmetric shift grid

$$\delta \in \{-1.00, -0.75, -0.50, -0.25, 0, 0.25, 0.50, 0.75, 1.00\},$$

so $\delta = 0.25$ corresponds to a 25 basis point move, and $H = 12$. For each horizon h , the LP-style outcome is the h -month cumulative (excess) return defined in Eq. (21). As is standard with forward cumulative outcomes, we drop the last H observations to avoid missing forward sums.

Nuisance estimation and reuse across horizons. The estimates in Table 8 are generated by the ScoreMatchingRiesz pipeline:

- A **single** data score network is fit once on the regressor process $\{X_t = (D_t, Z_t)\}$. In code, the Data-ScoreMatchingRiesz model is an MLP with two hidden layers (64,64) and ELU activations, trained for 4,000 steps with minibatch size 256. Gaussian noise is injected only in the (standardized) policy coordinate with $\sigma \sim \text{Unif}[0.05, 0.5]$, and the weighting uses $\lambda(\sigma) = \sigma^2$.
- For each horizon h , a separate outcome regression $\hat{\gamma}_h(x)$ is trained on the full sample with an MLP (two hidden layers (128,128), ELU activations, 2,000 epochs). The derivative $\partial_d \hat{\gamma}_h(X_t)$ used for AME is computed by automatic differentiation.
- For each δ on the grid, the learned score is integrated numerically to produce $\widehat{\log r_\delta}(x)$ and hence $\hat{r}_\delta(x)$, using a trapezoidal rule with 64 integration steps. These ratios enter the orthogonal score for the one-sided shift effect $\theta_h(\delta, 0)$ via the correction term $\mathbb{E}[(r_\delta(X_t) - 1) \cdot (Y_{t,h} - \hat{\gamma}_h(X_t))]$.

Because the data score is trained only once and depends only on X_t , it is reused across all horizons h and all shifts δ , ensuring that the entire policy path surface $(\delta, h) \mapsto \hat{\theta}_h(\delta, 0)$ is generated coherently.

HAC inference. For each (δ, h) , the standard error in Table 8 is a Newey–West HAC standard error computed from the centered orthogonal-score summands (estimated influence-function proxies), using a horizon-dependent truncation rule (in the implementation, $L_h = \max\{6, h\}$ for monthly data). The reported intervals are pointwise 95% CIs and do not adjust for multiple testing across δ .

From one-sided paths to APEs and symmetric paths. While Table 8 reports $\hat{\theta}_h(\delta, 0)$, it directly yields economically interpretable contrasts. Any two-point APE is recovered by differencing rows:

$$\hat{\theta}_h^{\text{APE}}(\delta_1, \delta_2) = \hat{\theta}_h(\delta_1, 0) - \hat{\theta}_h(\delta_2, 0).$$

In particular, the “25bp hike vs. 25bp cut” contrast is $\hat{\theta}_h(0.25, 0) - \hat{\theta}_h(-0.25, 0)$. A symmetric path can likewise be formed by $\hat{\theta}_h(\delta) = \hat{\theta}_h(\delta, 0) - \hat{\theta}_h(-\delta, 0)$.

Q.9. Empirical Patterns in Table 8

We stress again that this appendix is *methodological*: $D_t = \Delta FF_t$ is an observed policy-rate change and is not claimed to be an exogenous monetary policy shock. Accordingly, we interpret Table 8 as documenting conditional predictive associations under the chosen controls, and we use it primarily to illustrate how the proposed pipeline produces a rich policy-path object with uncertainty quantification.

(i) **Basic implementation checks.** The $\delta = 0$ row is exactly zero at all horizons, as required by the definition of the one-sided shift effect $\theta_h(0, 0) = 0$ and as a quick diagnostic that the table entries align with the intended target.

(ii) **Horizon-dependent sign patterns and a sign reversal around $h = 4$.** A striking feature of Table 8 is that the sign of $\hat{\theta}_h(\delta, 0)$ depends strongly on the horizon. For short horizons $h = 2$ and $h = 3$, the estimates are positive for negative shifts and negative for positive shifts. For example, at $h = 3$,

$$\hat{\theta}_3(-0.25, 0) = 0.0244 \quad \text{whereas} \quad \hat{\theta}_3(0.25, 0) = -0.0236,$$

and the corresponding CIs exclude zero. From $h = 4$ onward, the signs flip: negative shifts are associated with negative effects, and positive shifts with positive effects. For instance, at $h = 12$,

$$\hat{\theta}_{12}(-0.25, 0) = -0.0219 \quad \text{whereas} \quad \hat{\theta}_{12}(0.25, 0) = 0.0256,$$

again with tight CIs.

This horizon-dependent sign reversal is consistent with the economic fact that policy-rate changes are endogenous to the macro-financial state: rate cuts tend to occur in deteriorating conditions (which also predict lower medium-run equity returns), while rate hikes tend to occur in strengthening conditions (which predict higher medium-run returns). Even with rich controls and lags, residual “information effects” and policy endogeneity can dominate the medium-run predictive patterns. From the methodological perspective, the key point is that the policy path makes these sign changes visible rather than compressing them into a single linear coefficient.

(iii) **APE contrasts: “hike vs. cut” flips sign with horizon.** Because Table 8 is one-sided, the finite-shift APE for a 25bp hike versus a 25bp cut is

$$\hat{\theta}_h(0.25, 0) - \hat{\theta}_h(-0.25, 0).$$

Using the table entries, this contrast is negative at short horizons and positive at longer horizons. For example:

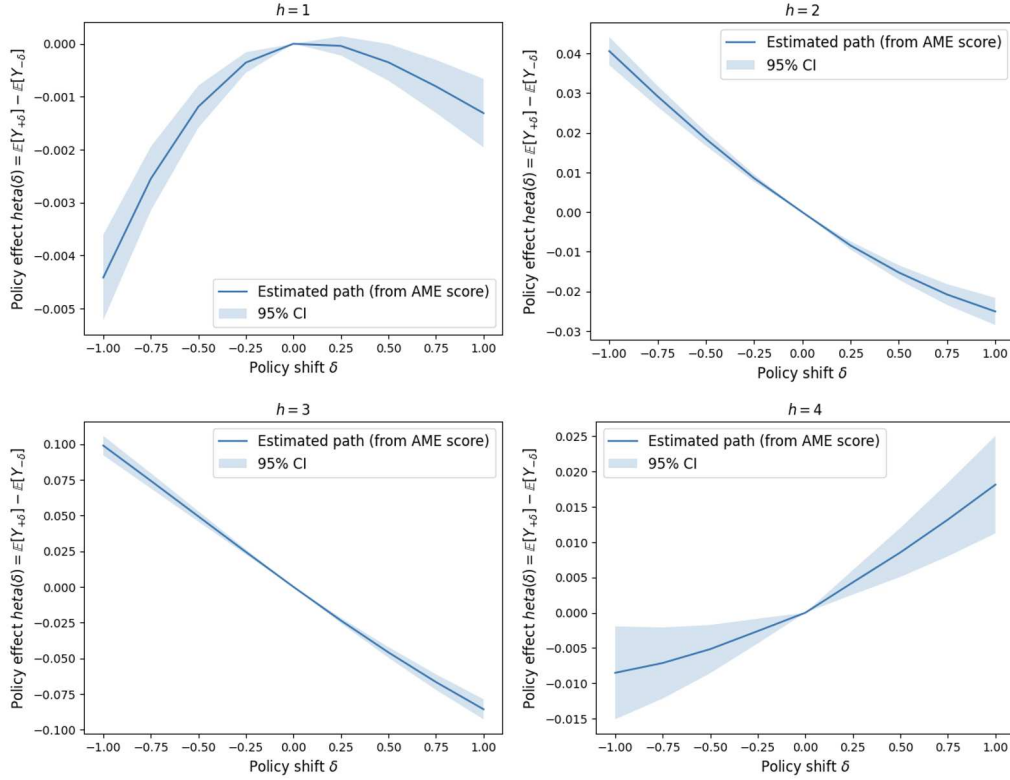
$$h = 2: -0.0084 - 0.0086 \approx -0.0170, \quad h = 12: 0.0256 - (-0.0219) \approx 0.0475.$$

Thus, the same policy magnitude can be associated with opposite-signed cumulative return changes depending on the horizon, underscoring why reporting an entire δ - h surface is informative in macro-finance settings.

(iv) **Approximate linearity in δ and asymmetry between easing and tightening.** Within many horizons, $\hat{\theta}_h(\delta, 0)$ scales roughly proportionally with δ , suggesting that in the range $|\delta| \leq 1$ the estimated policy path is close to linear. For example, at $h = 12$ the positive-shift entries increase nearly linearly: 0.0256 (25bp), 0.0536 (50bp), 0.0836 (75bp), 0.1149 (100bp). However, the path is not perfectly odd or symmetric: the magnitude for $\delta = 1.00$ is 0.1149, while for $\delta = -1.00$ it is -0.0643 . This asymmetry is economically plausible (policy easing episodes can occur in more adverse states than tightening episodes) and statistically expected when the shifted distributions p_δ and $p_{-\delta}$ have different degrees of overlap with the baseline. We plot the policy paths in Figures 4–6.

(v) **Precision and where uncertainty grows.** Standard errors are small relative to effect magnitudes for most horizons and nonzero shifts, and most pointwise CIs exclude zero, especially for $|\delta| \geq 0.25$ and $h \geq 2$. At the same time, uncertainty generally increases with both horizon and $|\delta|$, which is consistent with: (i) stronger serial dependence and accumulated noise in h -month cumulative outcomes, and (ii) the increasing difficulty of extrapolation for larger distributional shifts. From a representer perspective, this is precisely the region where density-ratio-based corrections can become heavy-tailed; the table therefore doubles as a practical diagnostic of which parts of the policy path are empirically well-supported.

Takeaway for the methodological objective. Overall, Table 8 illustrates the central claim of ScoreMatchingRiesz in a real-data setting: a single learned treatment score, reused across horizons, delivers a full policy-path object with coherent pointwise HAC uncertainty. The resulting δ - h surface reveals horizon-dependent sign changes, near-linear scaling with δ , and asymmetries between tightening and easing—all of which would be obscured by reporting only an AME or a single APE.

Figure 4. Policy path with $h = 1, 2, 3, 4$.

R. IHDP Semi-Synthetic ATE Benchmark

This appendix reports an additional semi-synthetic ATE benchmark based on the IHDP data. The purpose of this experiment is to evaluate the ratio-based ATE construction using real covariates with known ground-truth potential-outcome means. Unlike the synthetic experiments in Section 9 and Appendix P, the covariates are not simulated by our data-generating process.

Data. We use the standard IHDP semi-synthetic benchmark with 100 training replications. For each replication, the observational training split contains $n = 672$ units and 25 covariates. Let $D_i \in \{0, 1\}$ denote the treatment indicator, $Z_i \in \mathbb{R}^{25}$ the covariate vector, and Y_i the observed factual outcome. The benchmark also provides noiseless potential-outcome means $\mu_0(Z_i)$ and $\mu_1(Z_i)$. For replication r , the ground-truth ATE is computed as

$$\theta_{0,r}^{\text{ATE}} = \frac{1}{n} \sum_{i=1}^n \{\mu_1(Z_i) - \mu_0(Z_i)\}.$$

Estimators. We compare five estimators. Time-SMR estimates the ratio components of the ATE Riesz representer by learning time scores and integrating them along bridges. Joint-SMR uses the joint-training variant that learns the data-score and time-score components with shared representation. SQ-Riesz is a squared-loss direct ratio-fitting baseline, corresponding to an LSIF-type Riesz-regression objective. BKL-Riesz is a logistic density-ratio estimation baseline. MLE is the standard augmented inverse probability weighting estimator using a logistic propensity-score model. For all neural-network-based methods, the outcome regression is fitted on (D, Z) , while the ratio models are fitted on Z . All methods use two-fold cross-fitting.

ATE representer construction. Let $\pi_0 = \mathbb{P}(D = 1)$ and let p_Z , $p_{Z|D=1}$, and $p_{Z|D=0}$ denote the marginal and treatment-conditional covariate densities. The ATE Riesz representer can be written as

$$\alpha_0^{\text{ATE}}(D, Z) = \frac{D}{e_0(Z)} - \frac{1-D}{1-e_0(Z)}, \quad e_0(Z) = \mathbb{P}(D = 1 \mid Z).$$

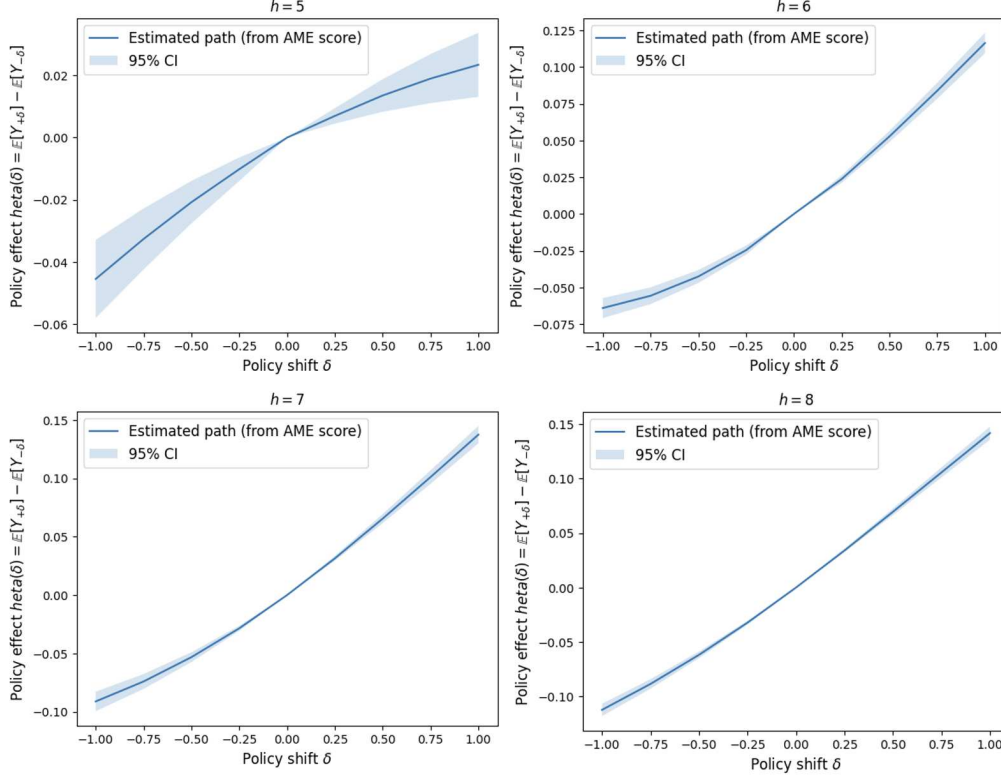


Figure 5. Policy path with $h = 5, 6, 7, 8$.

Equivalently,

$$\frac{1}{e_0(Z)} = \frac{p_Z(Z)}{\pi_0 p_{Z|D=1}(Z)}, \quad \frac{1}{1 - e_0(Z)} = \frac{p_Z(Z)}{(1 - \pi_0) p_{Z|D=0}(Z)}.$$

Thus, the ratio-based methods estimate $p_Z/p_{Z|D=1}$ and $p_Z/p_{Z|D=0}$ and form

$$\hat{\alpha}^{\text{ATE}}(D, Z) = D \frac{\hat{r}_1(Z)}{\hat{\pi}} - (1 - D) \frac{\hat{r}_0(Z)}{1 - \hat{\pi}},$$

where $\hat{r}_1$ estimates $p_Z/p_{Z|D=1}$, $\hat{r}_0$ estimates $p_Z/p_{Z|D=0}$, and $\hat{\pi}$ is the treatment fraction in the training fold.

Evaluation. For each replication r , we compute the cross-fitted orthogonal estimator

$$\hat{\theta}_r = \frac{1}{n} \sum_{i=1}^n \{ \hat{\gamma}(1, Z_i) - \hat{\gamma}(0, Z_i) + \hat{\alpha}^{\text{ATE}}(D_i, Z_i) (Y_i - \hat{\gamma}(D_i, Z_i)) \}.$$

We report signed bias, mean absolute error, RMSE, empirical coverage of nominal 95% confidence intervals, average standard error, and average confidence-interval length across 100 replications. The confidence intervals use the empirical variance of the cross-fitted orthogonal scores.

Results. Table 9 shows that the score-based ratio estimators perform favorably on this semi-synthetic benchmark. Joint-SMR has the smallest RMSE, 1.478, and Time-SMR has the second smallest RMSE, 1.571. The direct ratio-fitting baselines are less stable: SQ-Riesz has RMSE 27.954, BKL-Riesz has RMSE 84.072, and AIPW-logit has RMSE 19.643. The intervals for Time-SMR and Joint-SMR cover the truth in all 100 replications, with average lengths around 5.8, so their inference is conservative in this experiment. In contrast, the baselines have lower empirical coverage despite longer intervals in some cases, reflecting large finite-sample bias and unstable ratio estimation. These results support the use of time-score-based ratio estimation as a Riesz representer estimation module for ATE, but they should be interpreted as supplementary evidence rather than as evidence that one method is uniformly best across all designs.

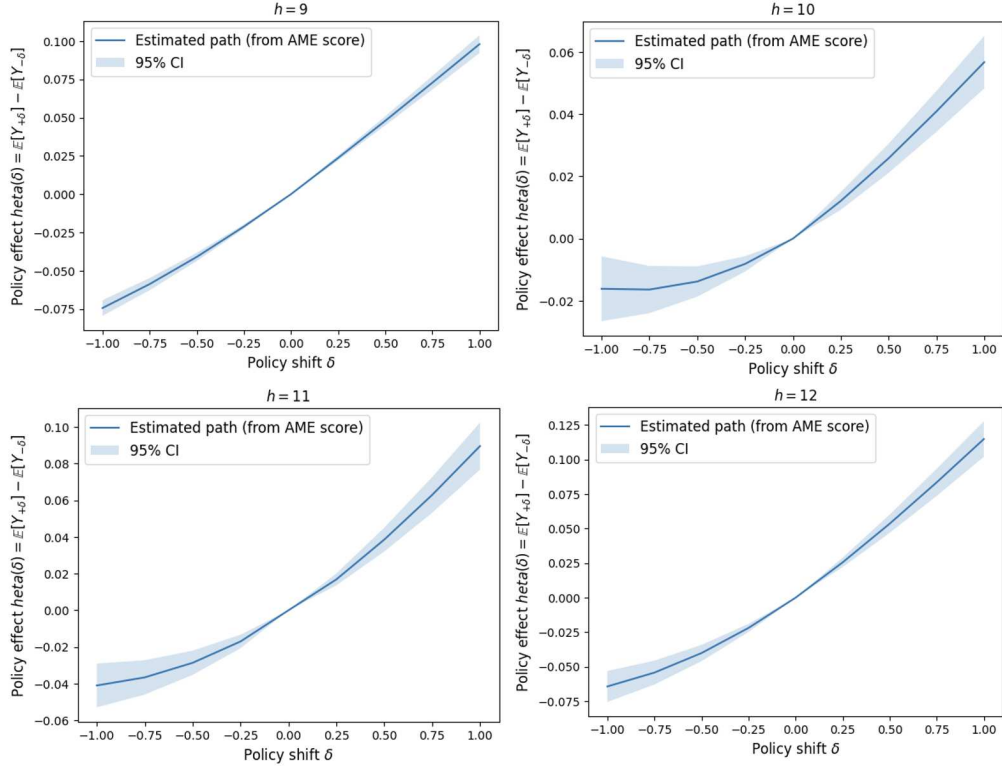
Figure 6. Policy path with $h = 9, 10, 11, 12$.

Table 9. IHDP semi-synthetic ATE benchmark over 100 replications.

Method	Estimate mean	Truth mean	Bias	MAE	RMSE	95% cov.	Avg. SE	Avg. length
Time-SMR	5.480	4.543	0.937	0.958	1.571	100.0%	1.479	5.799
Joint-SMR	5.404	4.543	0.860	0.885	1.478	100.0%	1.482	5.809
SQ-Riesz	23.654	4.543	19.111	19.111	27.954	0.0%	5.534	21.694
BKL-Riesz	52.838	4.543	48.294	48.294	84.072	19.0%	21.429	83.999
MLE	16.091	4.543	11.548	11.548	19.643	46.0%	5.906	23.151